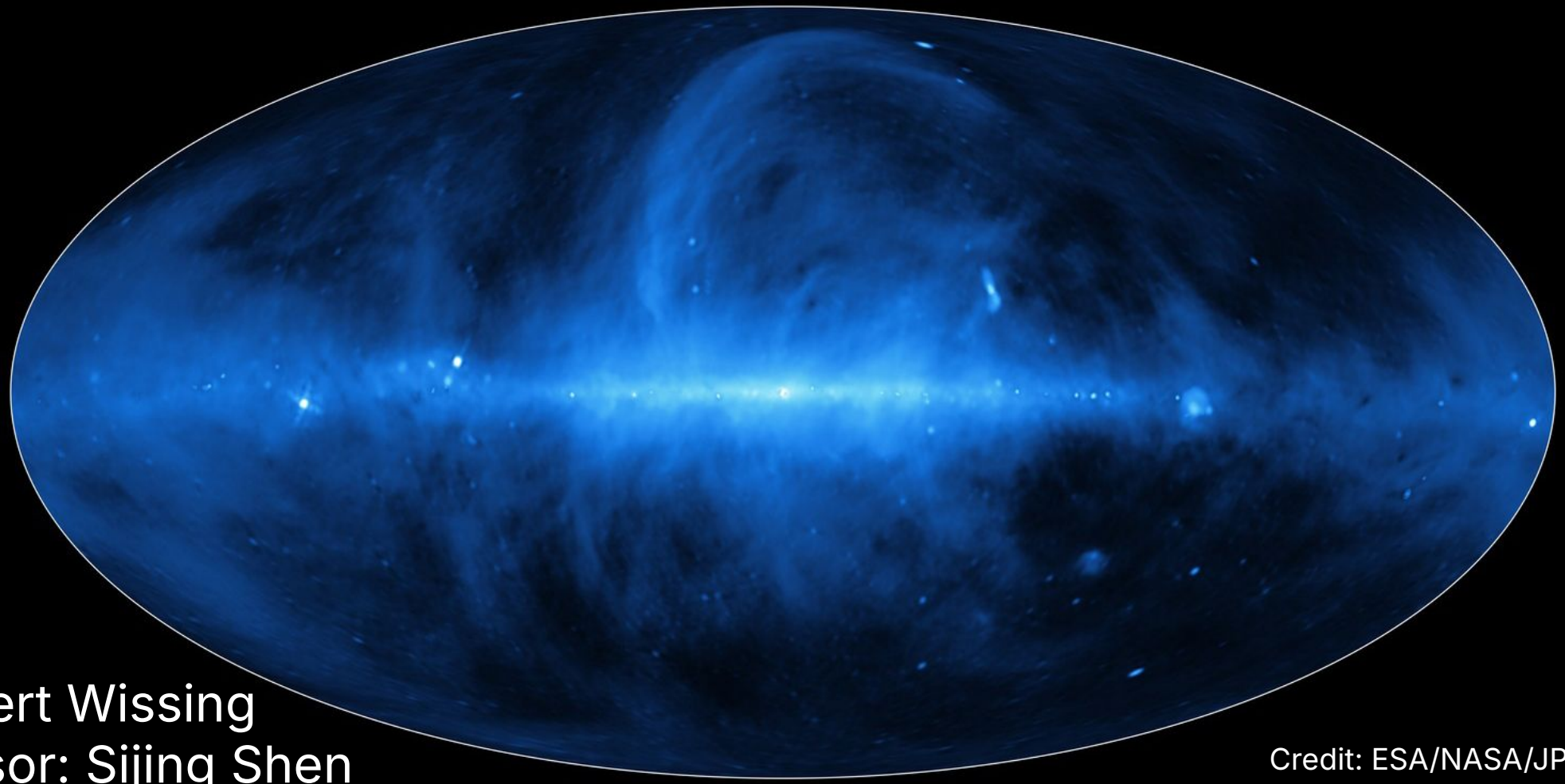
SPMHD with the geometric density force gradient operator (GDSPH)



Robert Wissing
Supervisor: Sijing Shen

Credit: ESA/NASA/JPL-Caltech

Magnetic fields in astrophysics

- Galaxy formation

Equipartition with turbulent energy – Dynamically important!
(Beck et al., 1996; Beck, 2005)

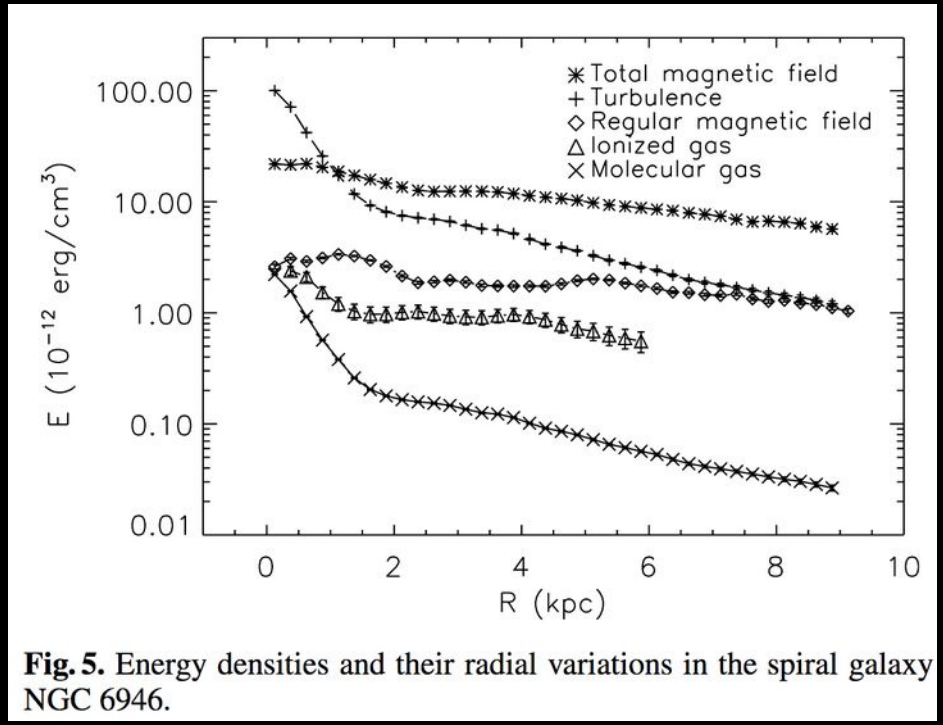


Fig. 5. Energy densities and their radial variations in the spiral galaxy NGC 6946.

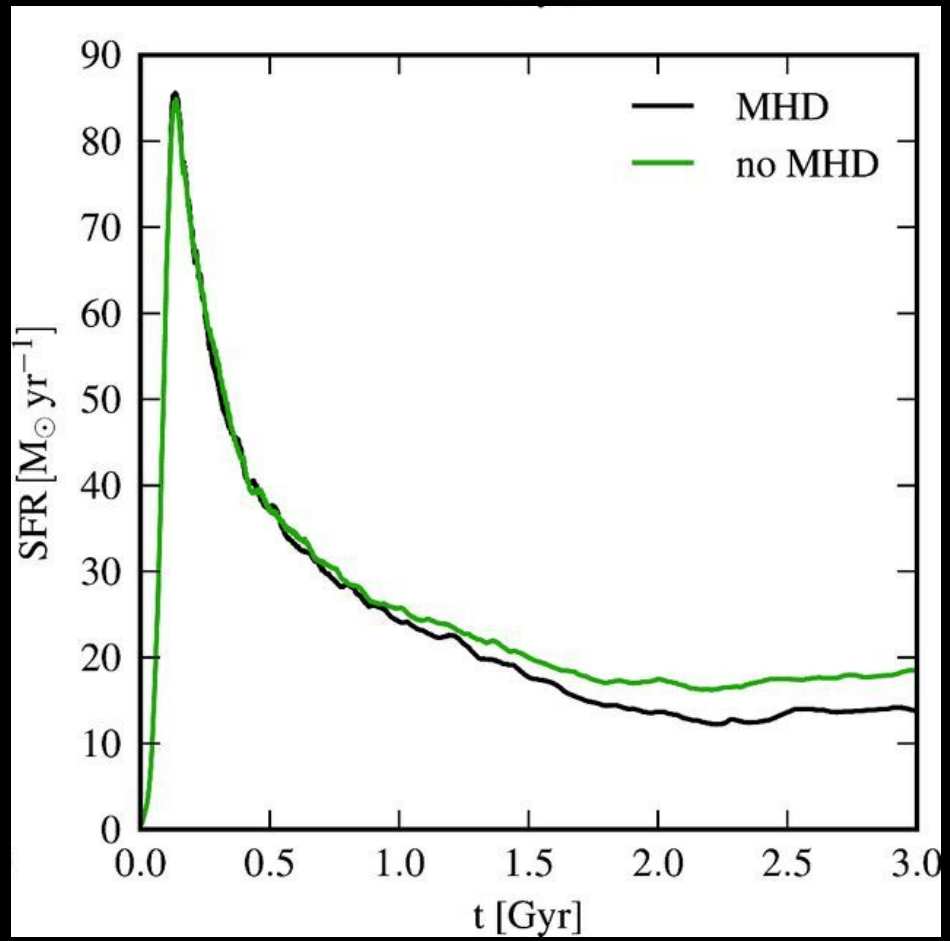
Beck (2007)

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Non-thermal pressure support - Suppresses star formation
(Wang and Abel, 2009; Pakmor and Springel , 2012)



Pakmor and Springel , 2012

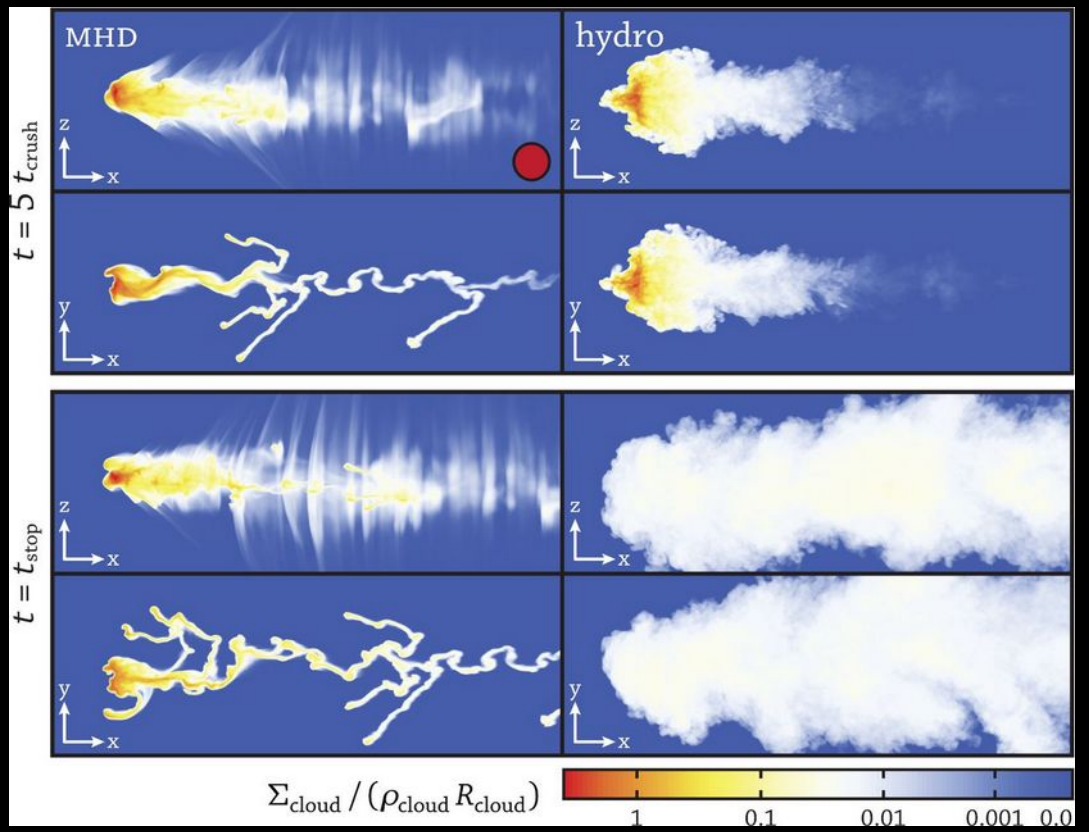
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Stabilizing effect on fluid instabilities (KH+RT)
(e.g., McCourt et al. (2015))



McCourt et al. (2015)

Magnetic fields in astrophysics

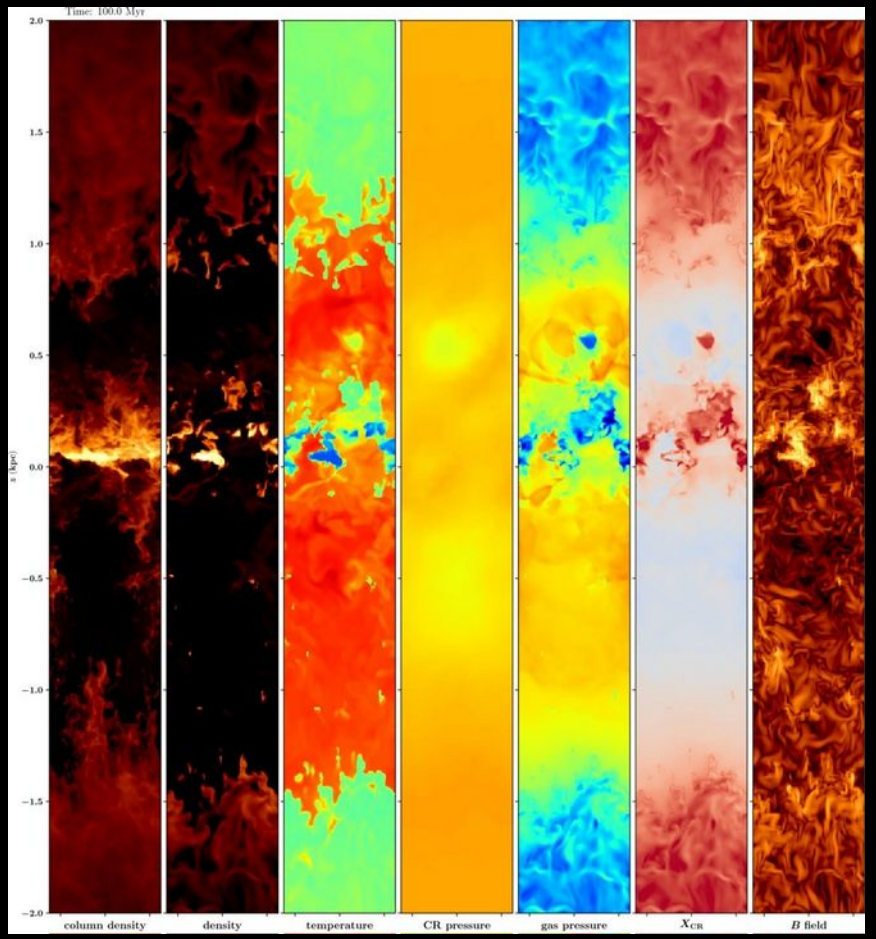
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Correct transport of CR - Galactic outflows
Girichidis et al. (2016a)



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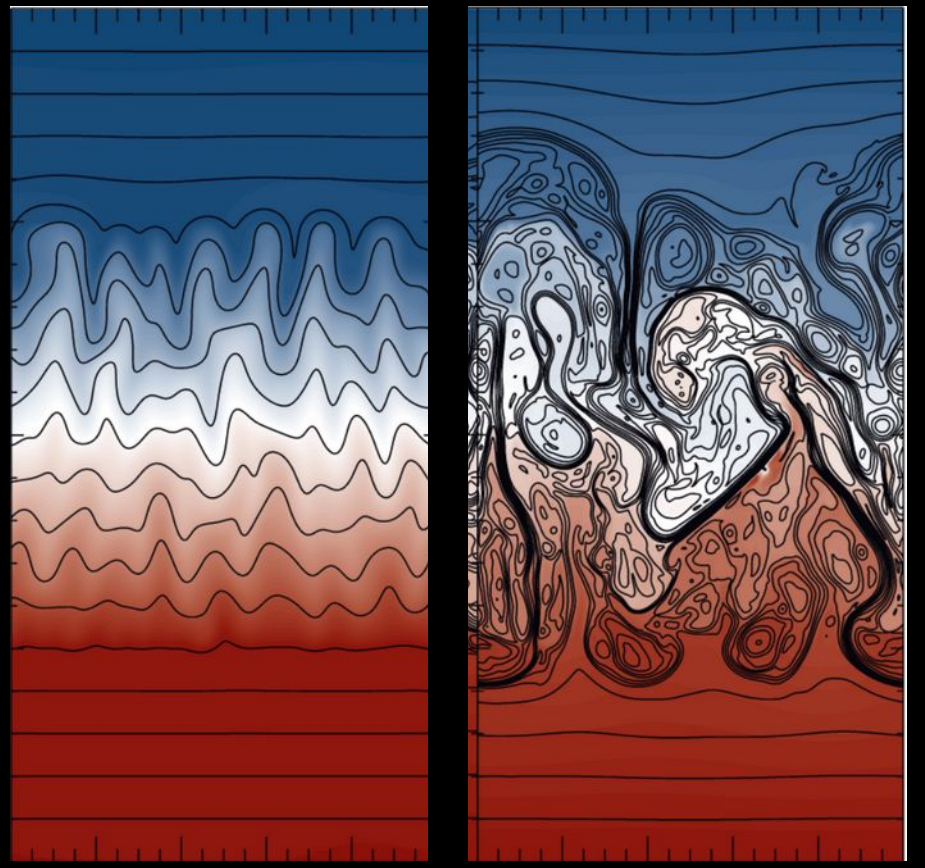
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Buoyancy instabilities - Intracluster medium (ICM)



Kunz et al. (2012)

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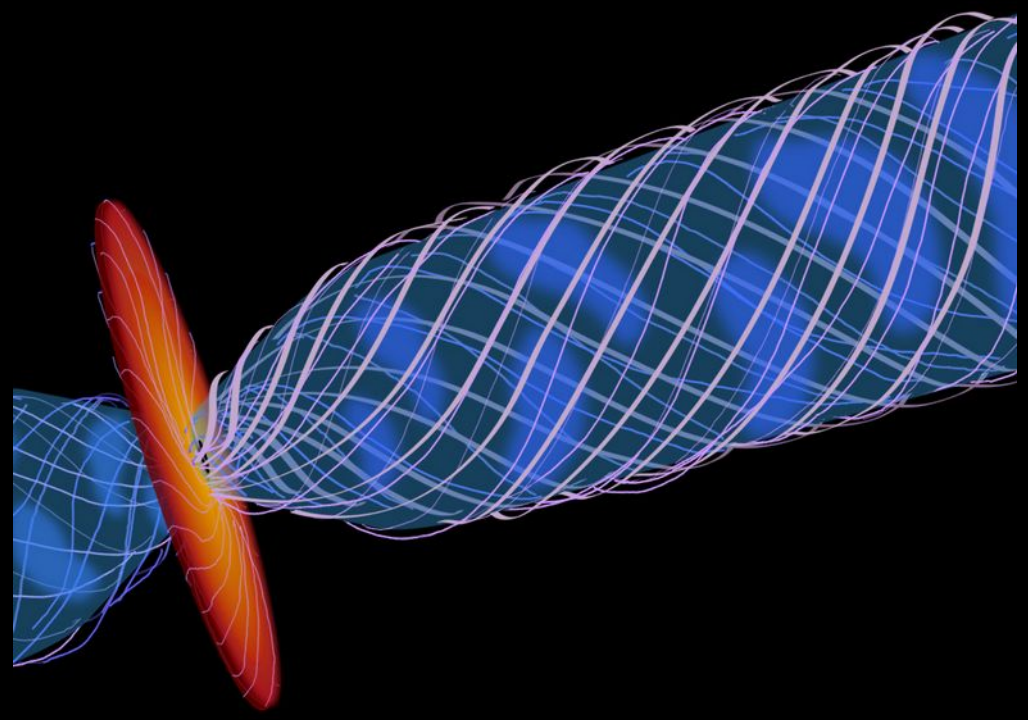
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- MHD instabilities

Magnetorotational instability - Ionized accretion disks

Buoyancy instabilities - Intracluster medium (ICM)

- Launching of astrophysical jets
- And much more...



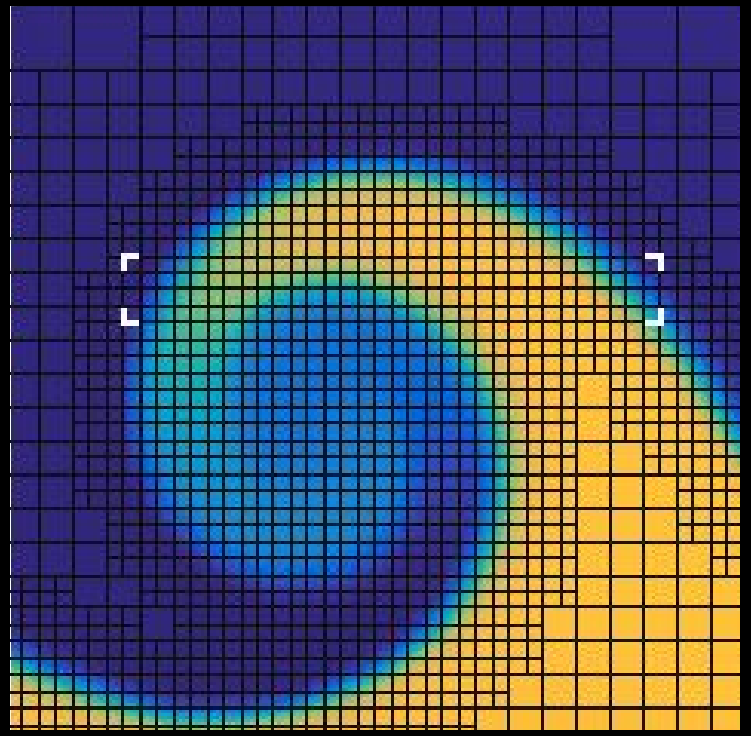
Credit: NASA

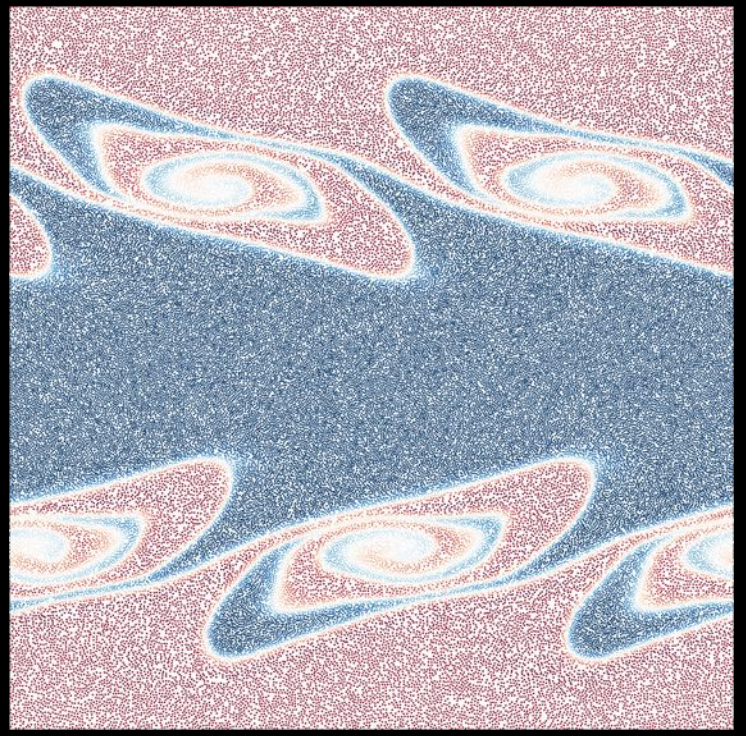
Numerical Methods

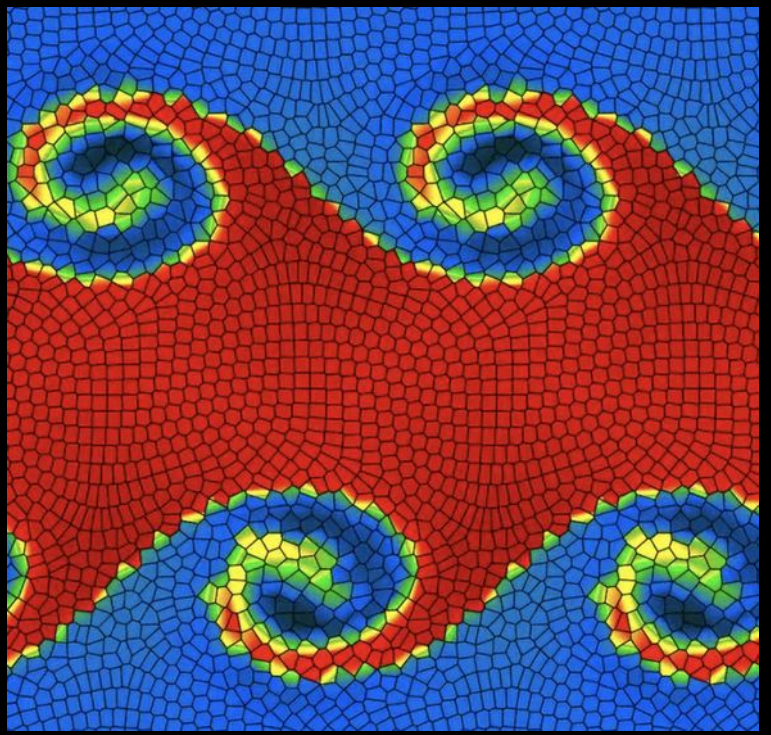
- Eulerian grid
- Smoothed particle hydrodynamics (SPH)
- Hybrids

Moving mesh

Meshless finite mass (MFM)







Smoothed particle magnetohydrodynamics (SPMHD)

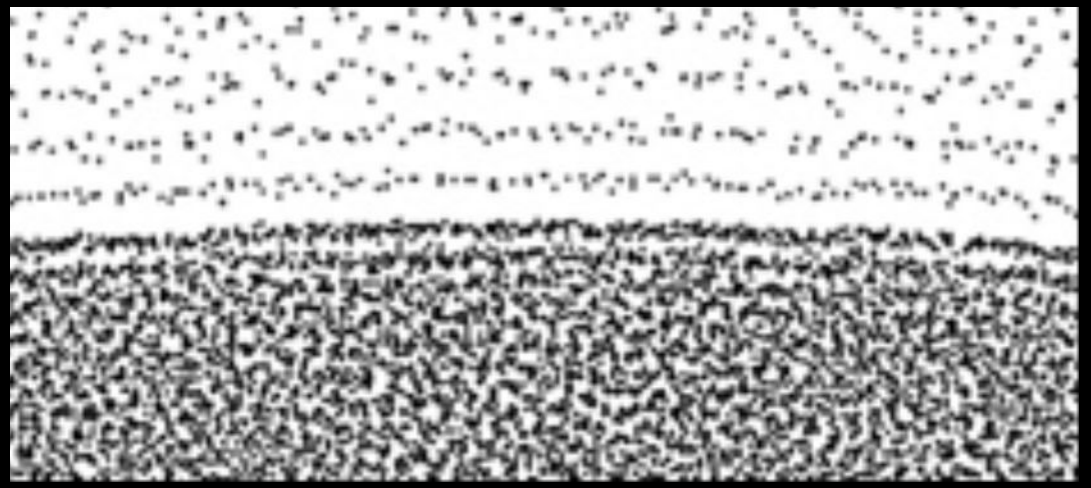
- Many improvements in recent years
- Wendland kernels improves convergence
  Dehnen & Aly 2012
- Modern switches for artificial viscosity and resistivity
  Wadsley et al. 2017; Price et al. 2017
- Better divergence cleaning
  Tricco et al. 2016

Fluid instabilites in SPH

- Standard SPH supresses fluid instabilities
  Agertz et al. 2007

Surface tension effect

Inconsistent force calculations of particles across contact discontinuities



Agertz et al. (2007)

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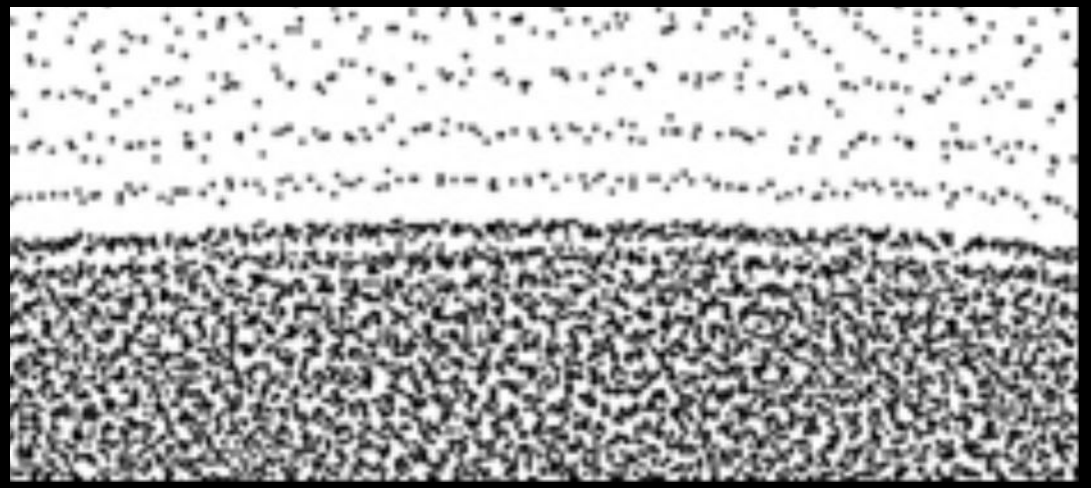
Inconsistent force calculations of particles across contact discontinuities

- Different solutions to the problem

Pressure/entropy SPH (PSPH)
Ritchie & Thomas 2001; Hopkins 2013

Artificial conductivity
Price 2008

New gradient operator
Wadsley et al. 2017



Agertz et al. (2007)

GDSPH gradient operator

- Different gradient operators

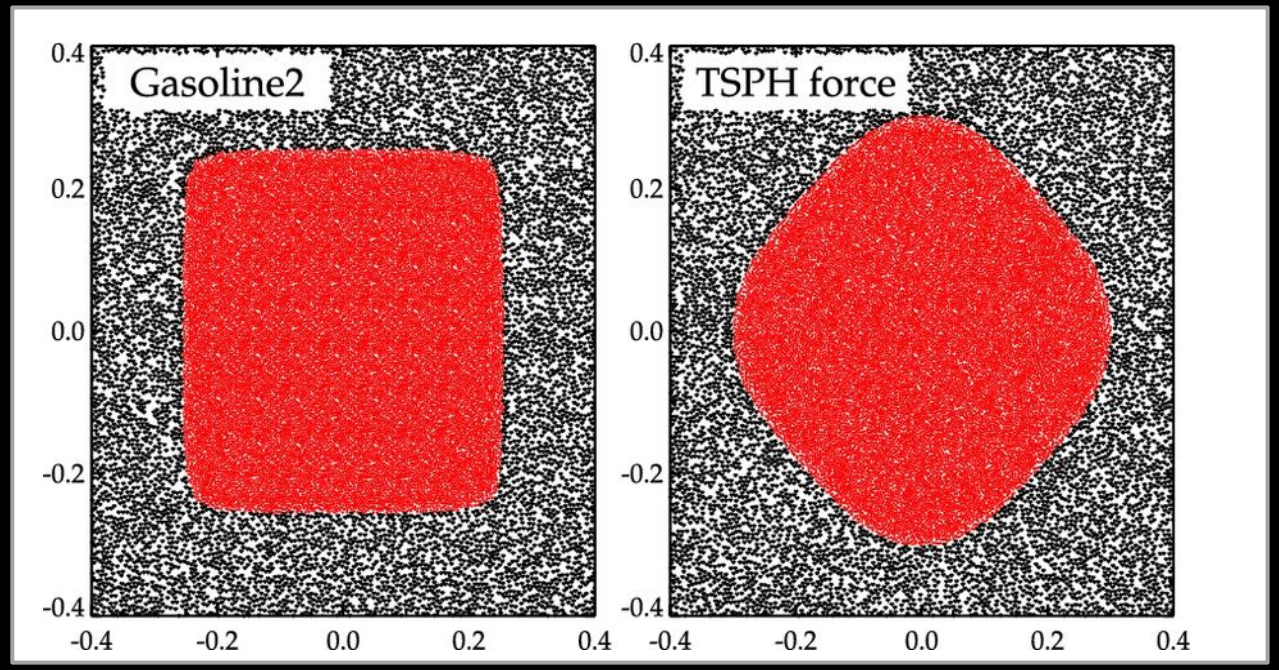
TSPH:

$$\frac{dv_a}{dt} = \frac{\nabla P}{\rho} \approx \sum_b m_b \left(\frac{P_a}{\rho_a^2} + \frac{P_b}{\rho_b^2} \right) \nabla_a \bar{W}_{ab}$$

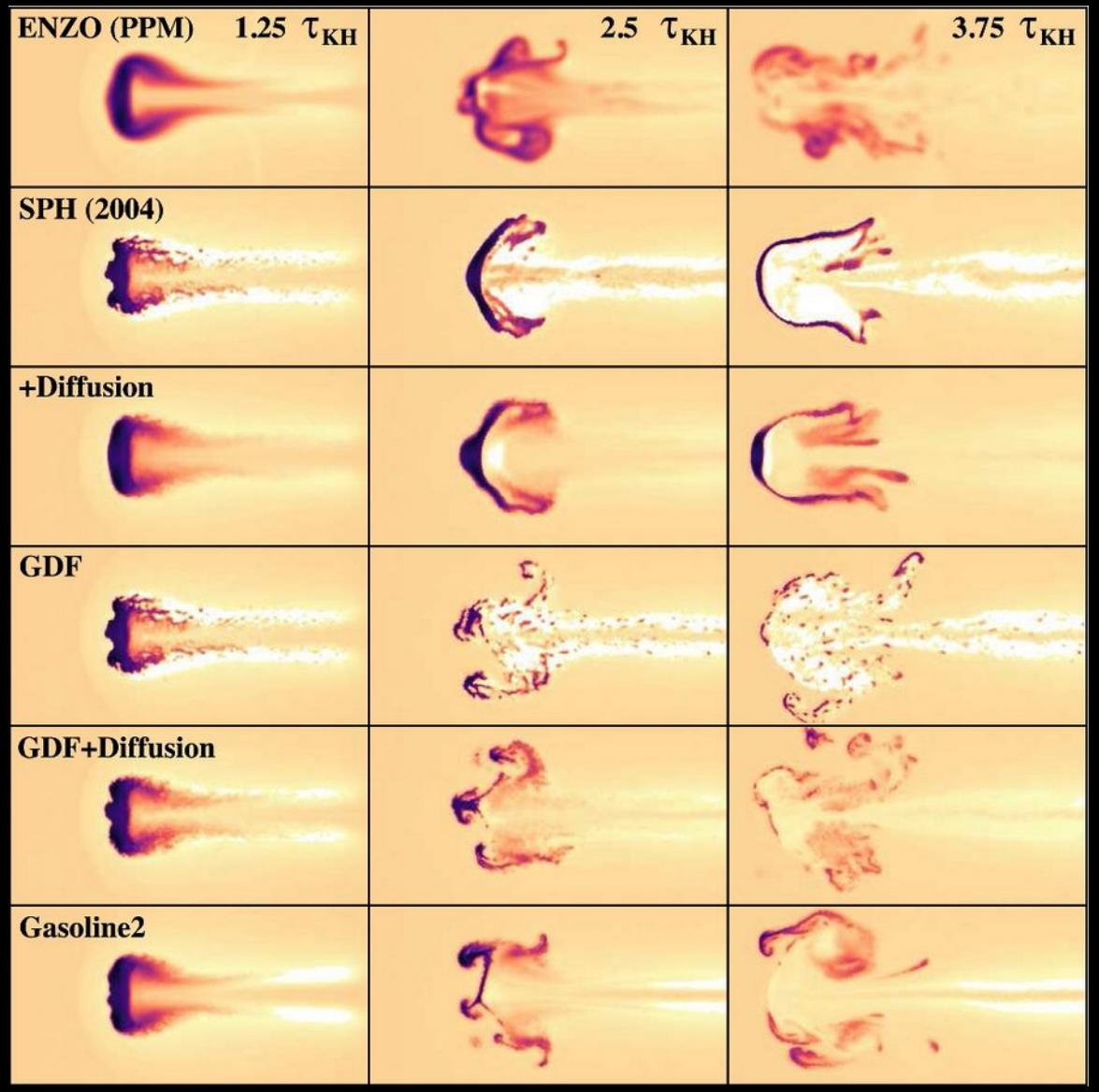
GDSPH:

$$\frac{dv_a}{dt} = \frac{\nabla P}{\rho} \approx \sum_b m_b \left(\frac{P_a + P_b}{\rho_a \rho_b} \right) \nabla_a \bar{W}_{ab}$$

- Significantly improves convergence for problems involving large density contrasts



Wadsley et al. 2017



Wadsley et al. 2017

Implementation of GDSPMHD

- MHD with GDSPH operator

Lorentz force: $\frac{dv_a}{dt} = -\sum_b \frac{m_b}{\rho_a \rho_b}\left[\frac{B_a^2+B_b^2}{2}\nabla_a \bar{W}_{ab} - B_a(B_a \cdot \nabla_a \bar{W}_{ab}) - B_b(B_b \cdot \nabla_a \bar{W}_{ab})\right]$

Induction: $\frac{dB_a}{dt} = \sum_b \frac{m_b}{\rho_b}[B_a(v_{ab} \cdot \nabla_a \bar{W}_{ab}) - v_{ab}(B_a \cdot \nabla_a \bar{W}_{ab})]$

- New artificial resistivity switch
  Price et al. 2017

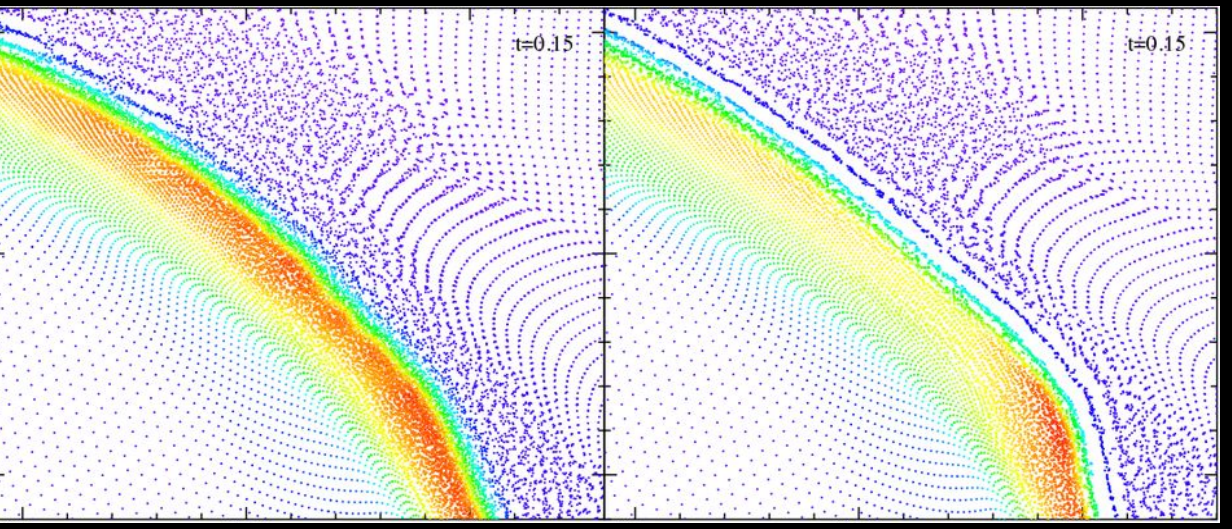
- Constrained hyperbolic divergence cleaning
  Tricco et al. 2016

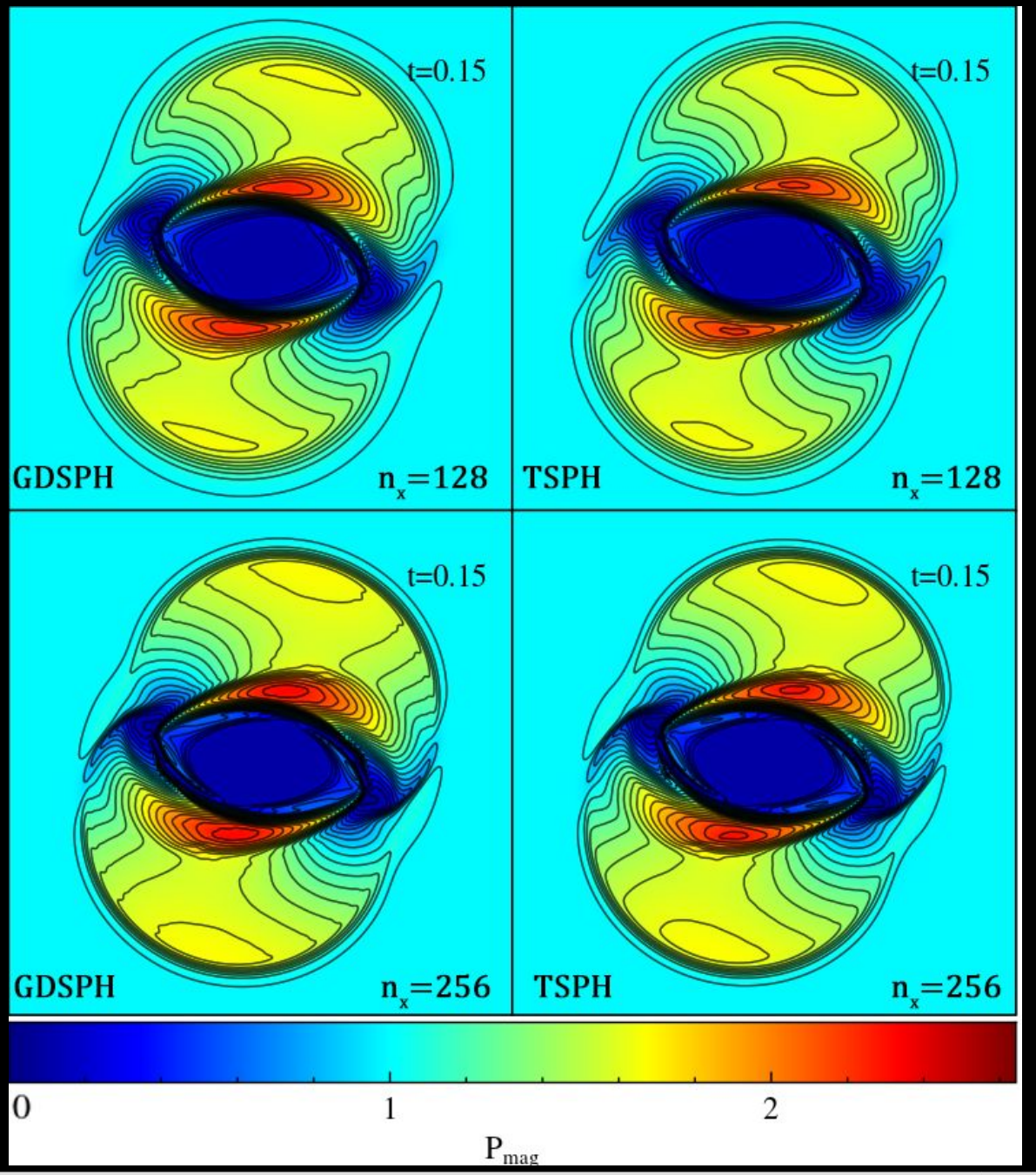
$$\frac{\partial^2(\nabla \cdot B)}{\partial t^2} - c_h^2 \nabla^2(\nabla \cdot B) + \frac{1}{\tau}\frac{\partial(\nabla \cdot B)}{\partial t} = 0$$

- Implemented in Gasoline2 and ChaNGa

Magnetic Rotor

GDSPH TSPH





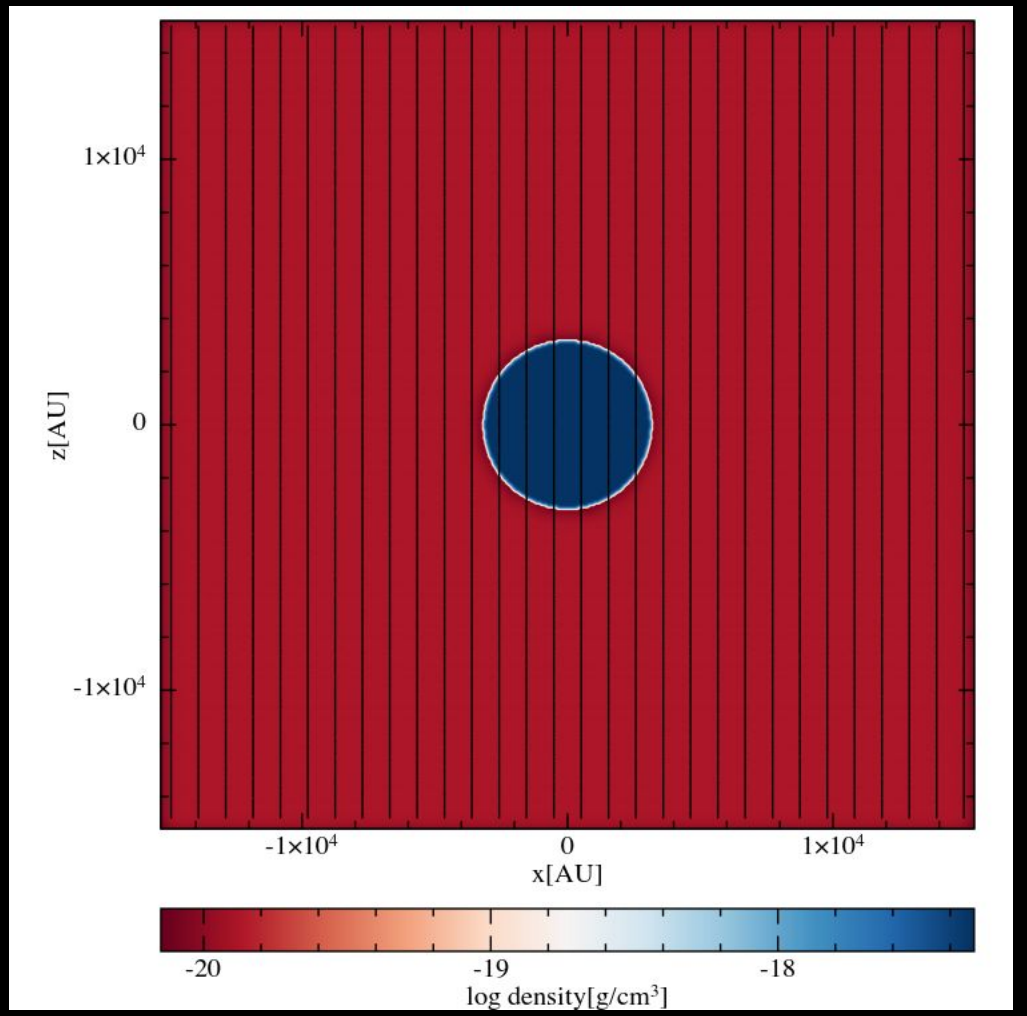
Magnetized core collapse

- Rotating, self-gravitating, magnitzed cloud
- Magnetic support against collapse

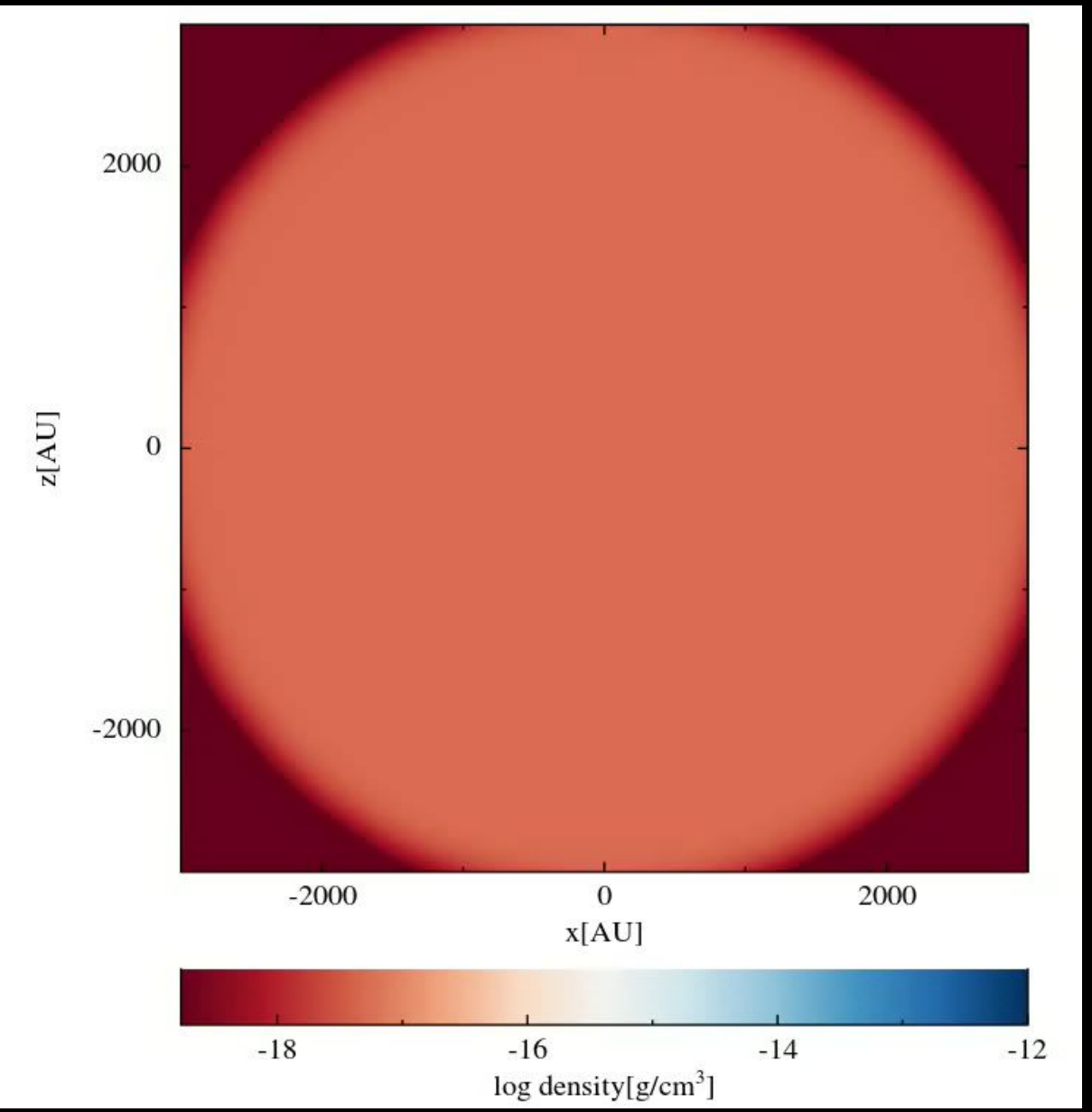
$$\mu = \frac{M}{\Phi} / \left(\frac{M}{\Phi}\right)_{crit}$$

Subcritical $\mu < 1$ Supercritical $\mu > 1$

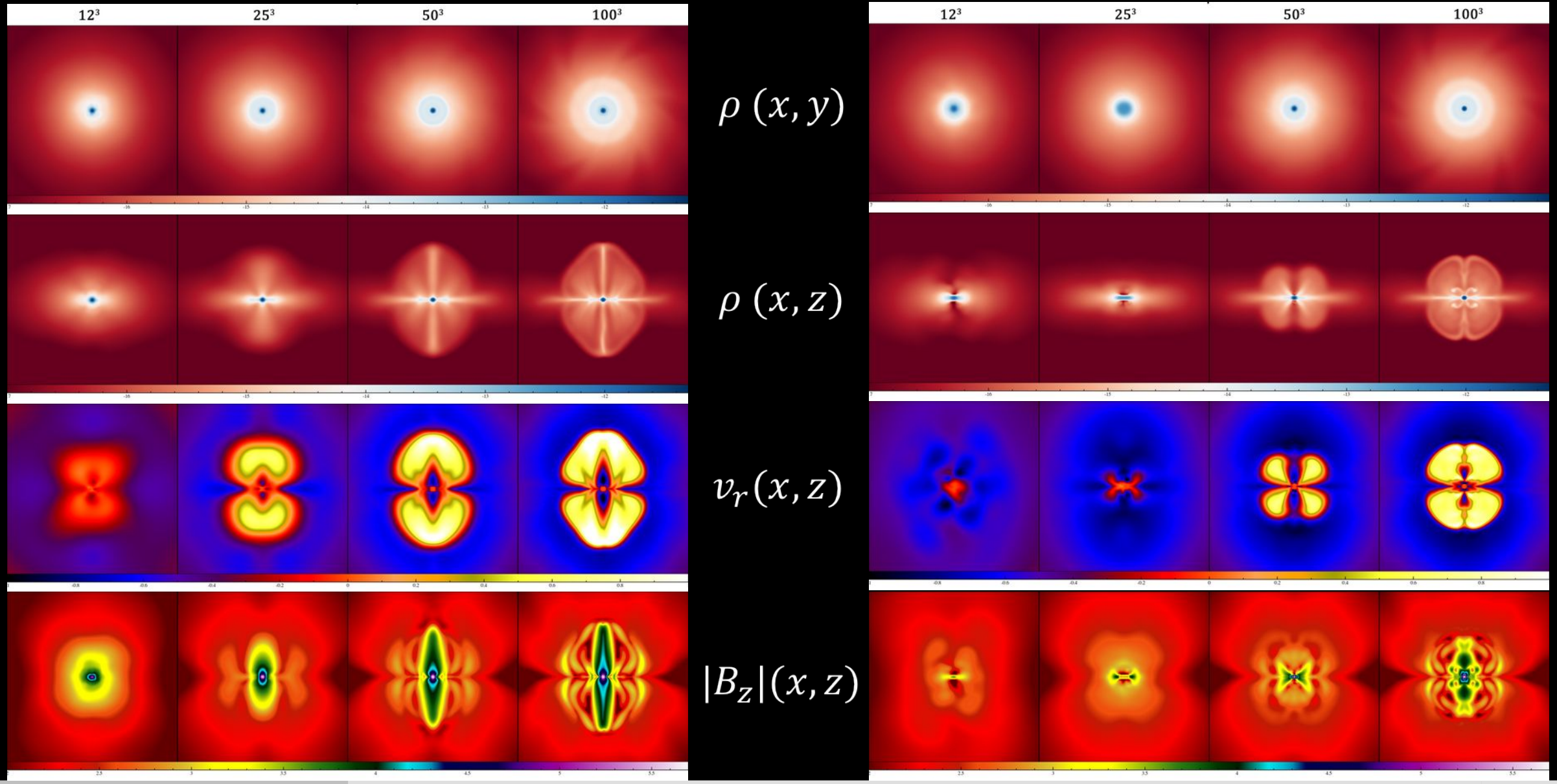
- For our set of initial conditions
  Hopkins & Raives (2016)
- $B_0 = \frac{610}{\mu}\hat{z} \quad [\mu G]$
- Barotropic EOS
- $\rho_{cloud} = 360 \times \rho_{out}$



GDSPH 50^3. $\mu = 10$



GDSPH Resolution study $\mu = 10$ TSPH



PSPH

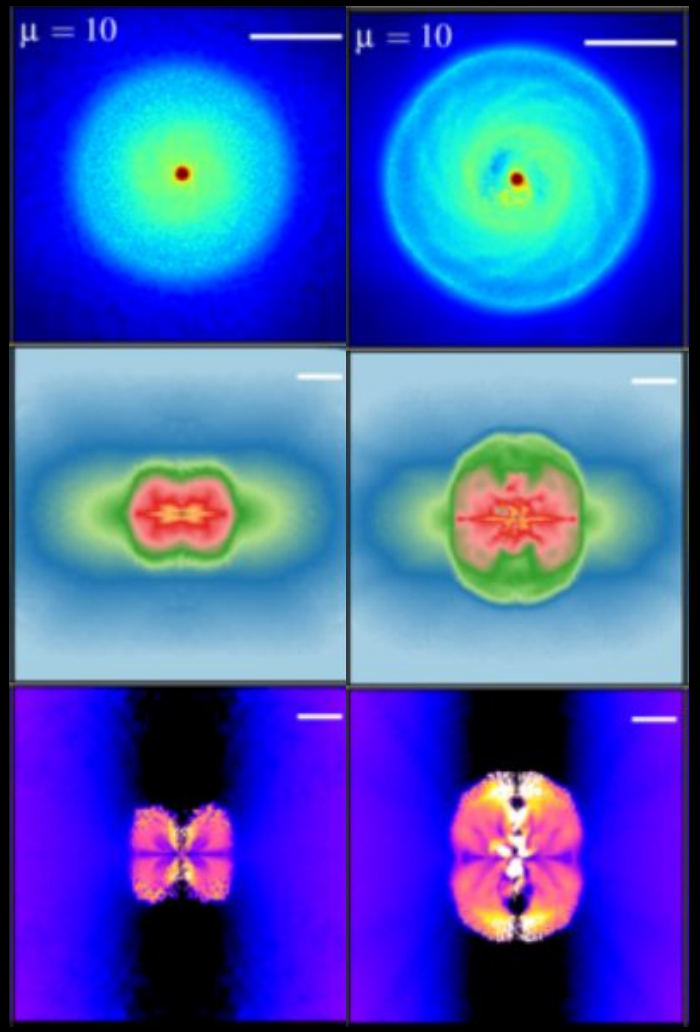
Hopkins & Raives (2016)

50^3 100^3

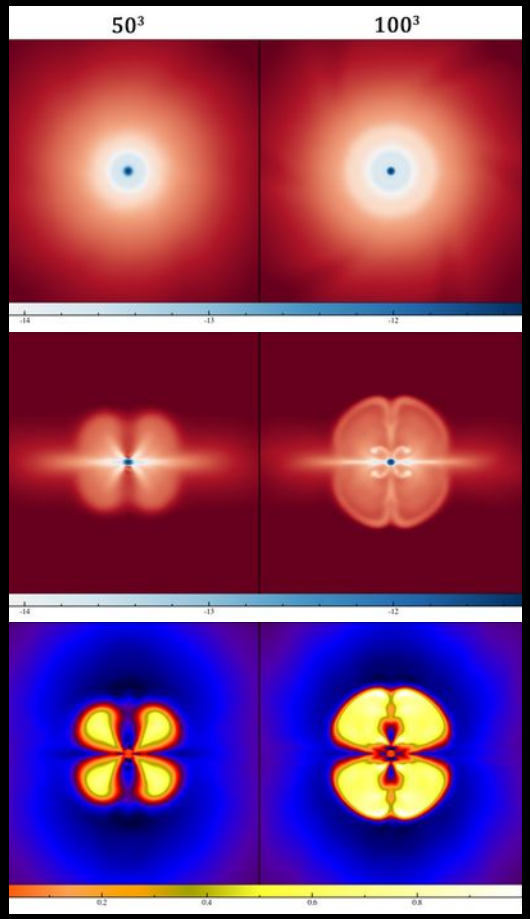
$\rho\ (x, y)$

$\rho\ (x, z)$

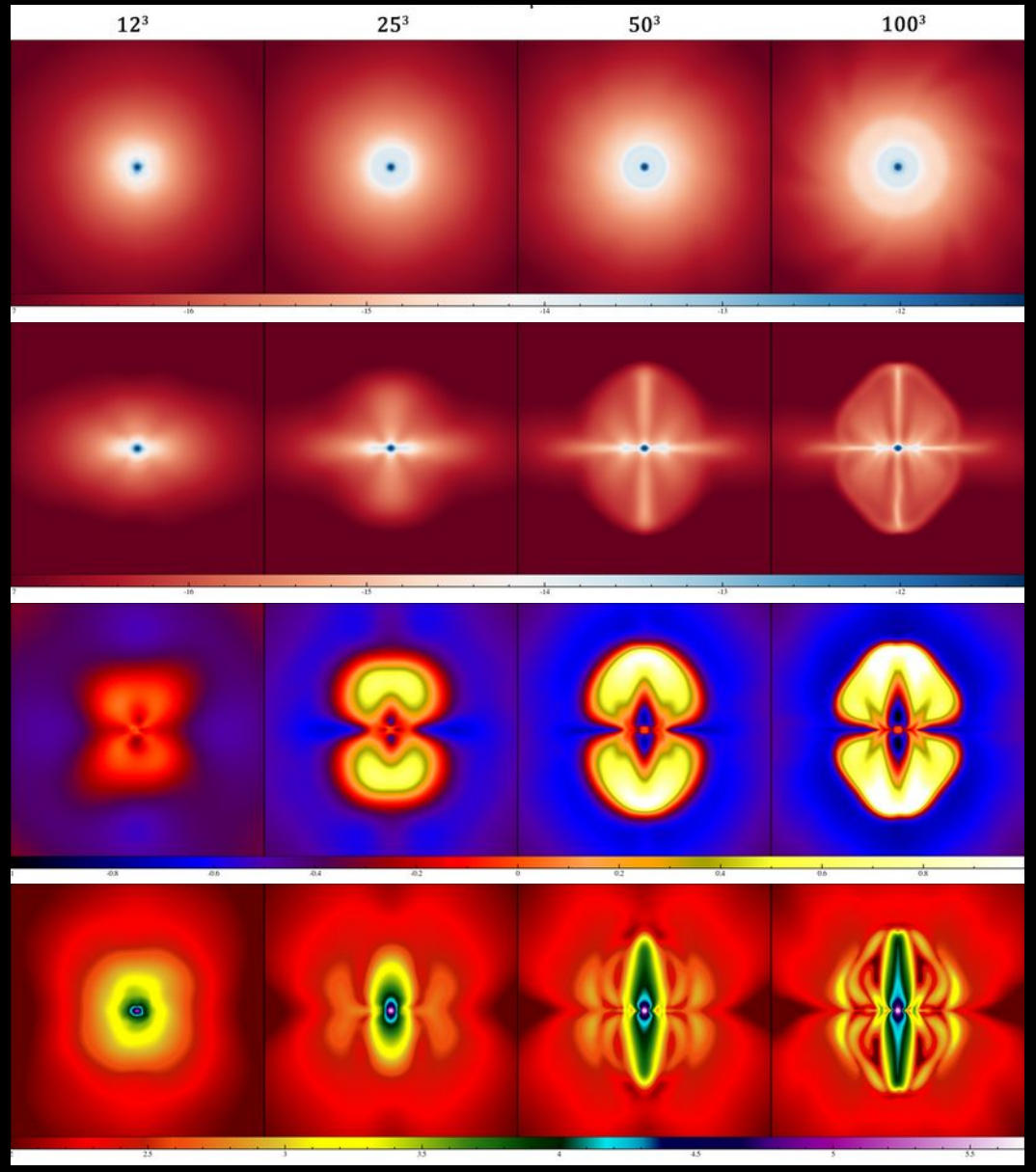
$v_r(x, z)$



TSPH

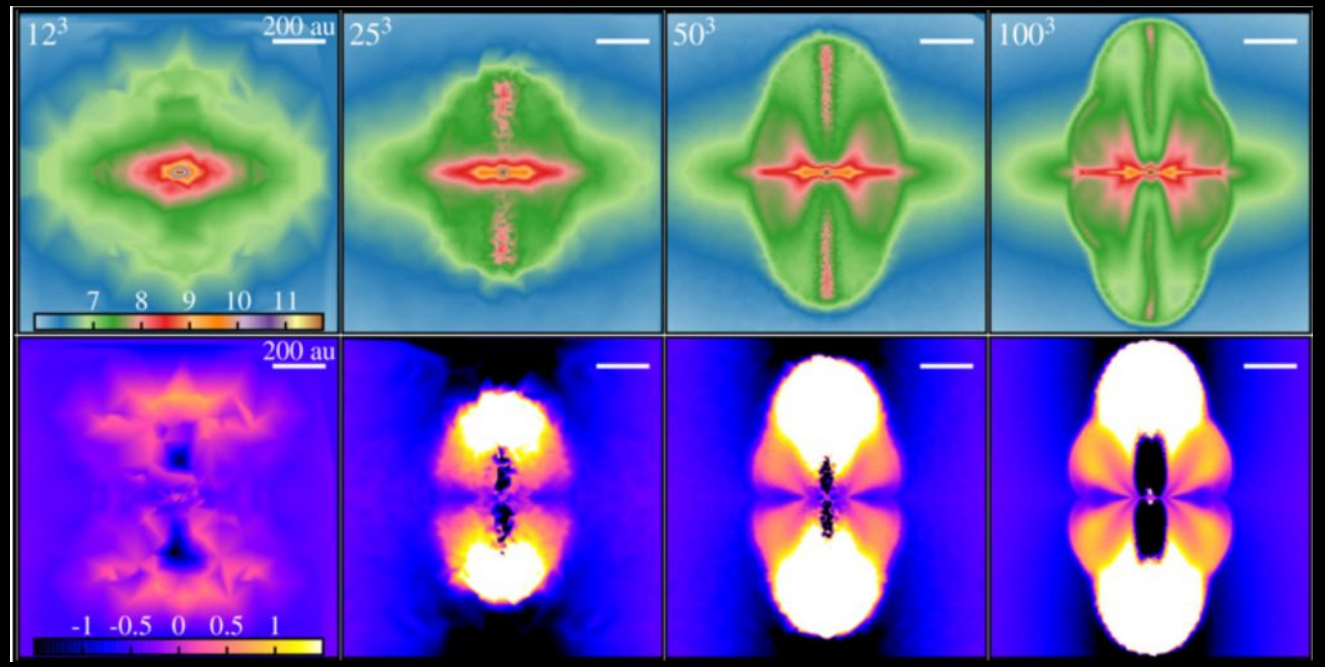


GDSPH

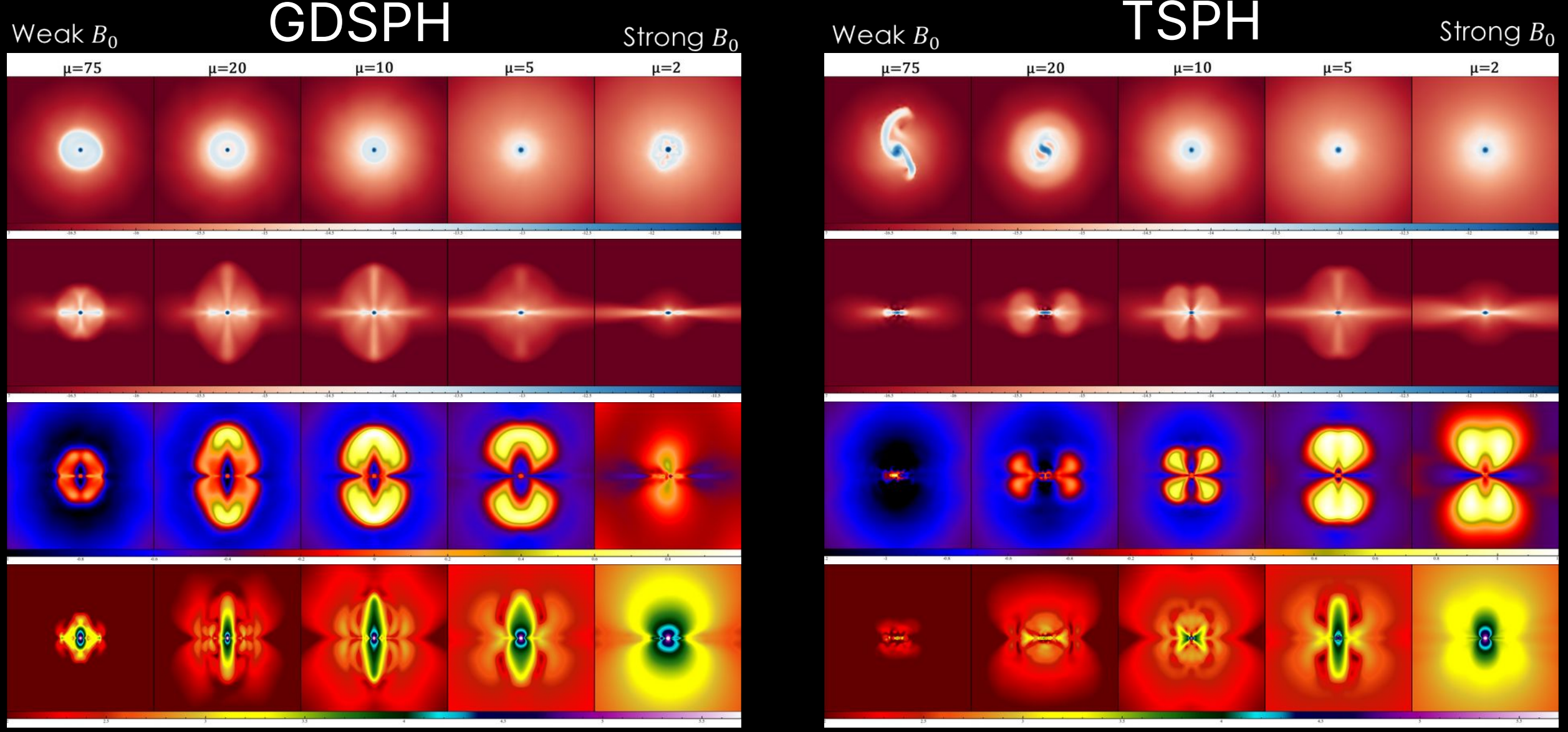


MFM

Hopkins & Raives (2016)



Varying μ – Resolution 50^3

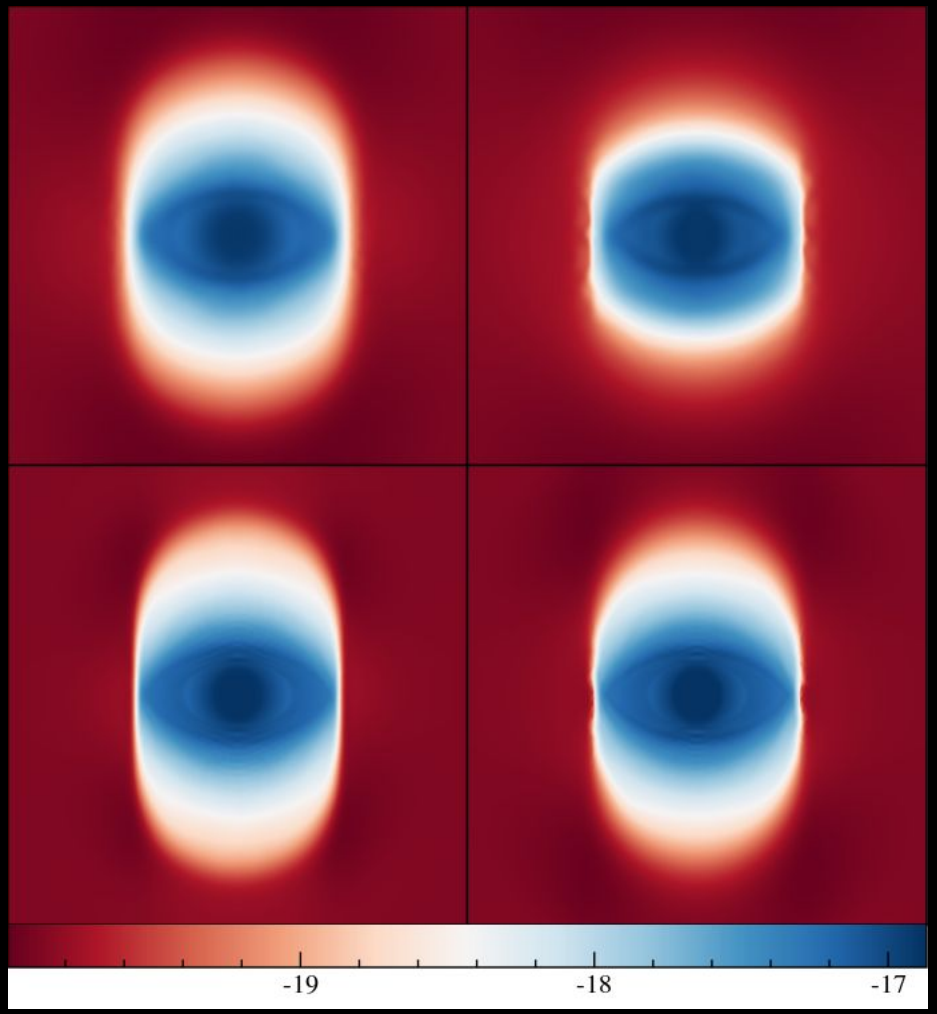


Cloud structure μ=2

GDSPH TSPH

50^3

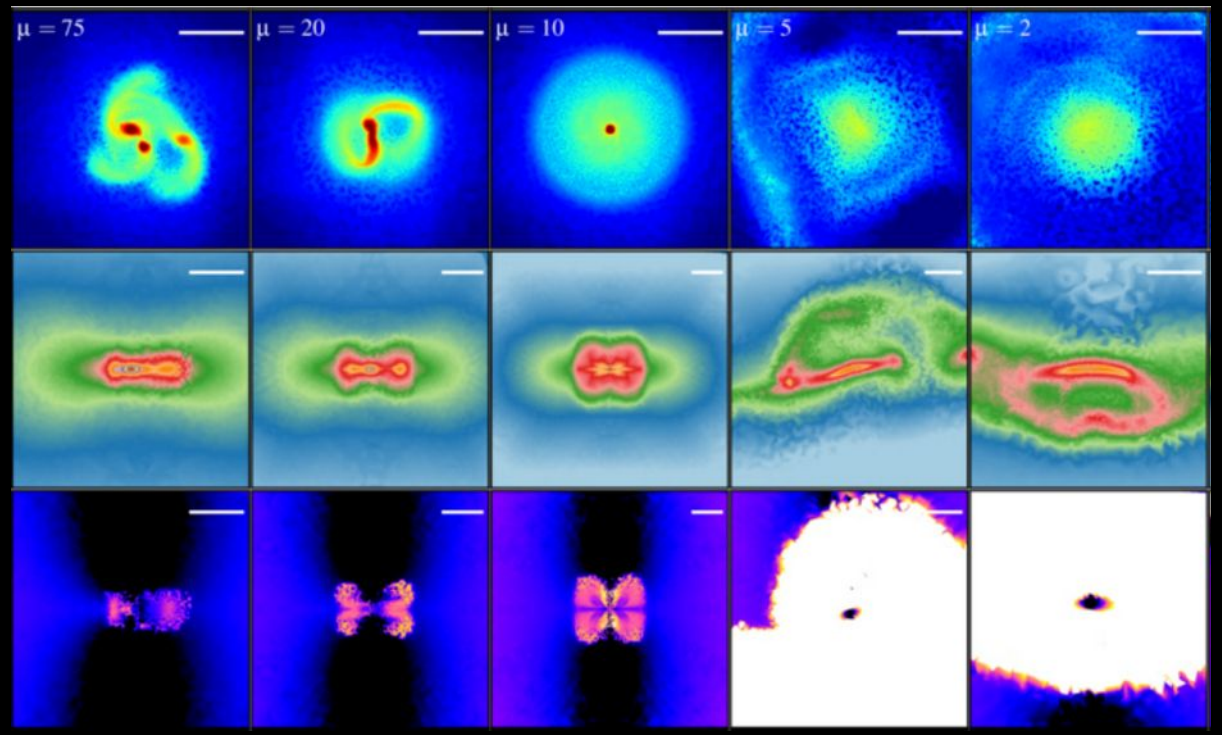
200^3



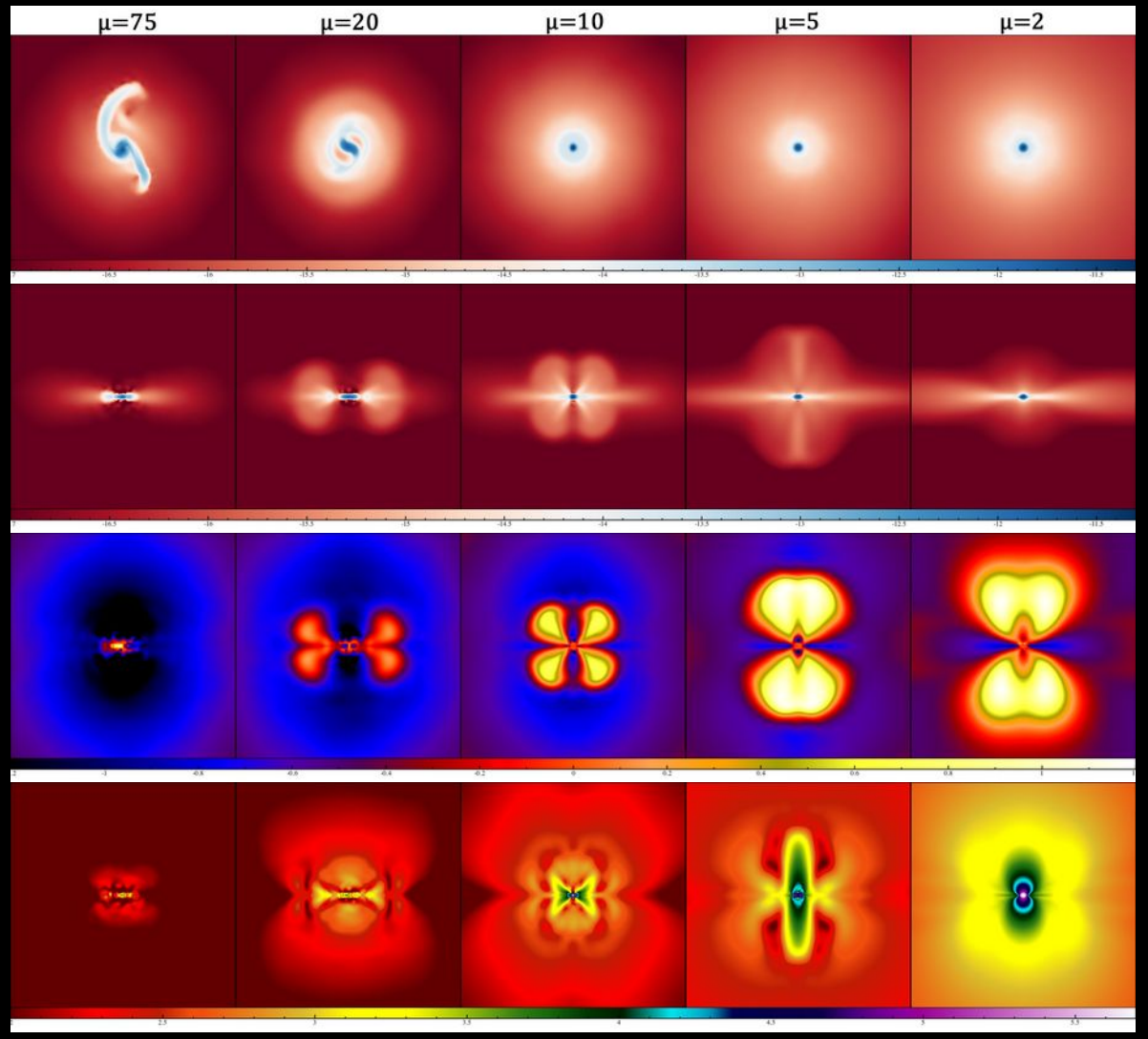
Varying μ – Resolution 50^3

PSPH

Hopkins & Raives (2016)

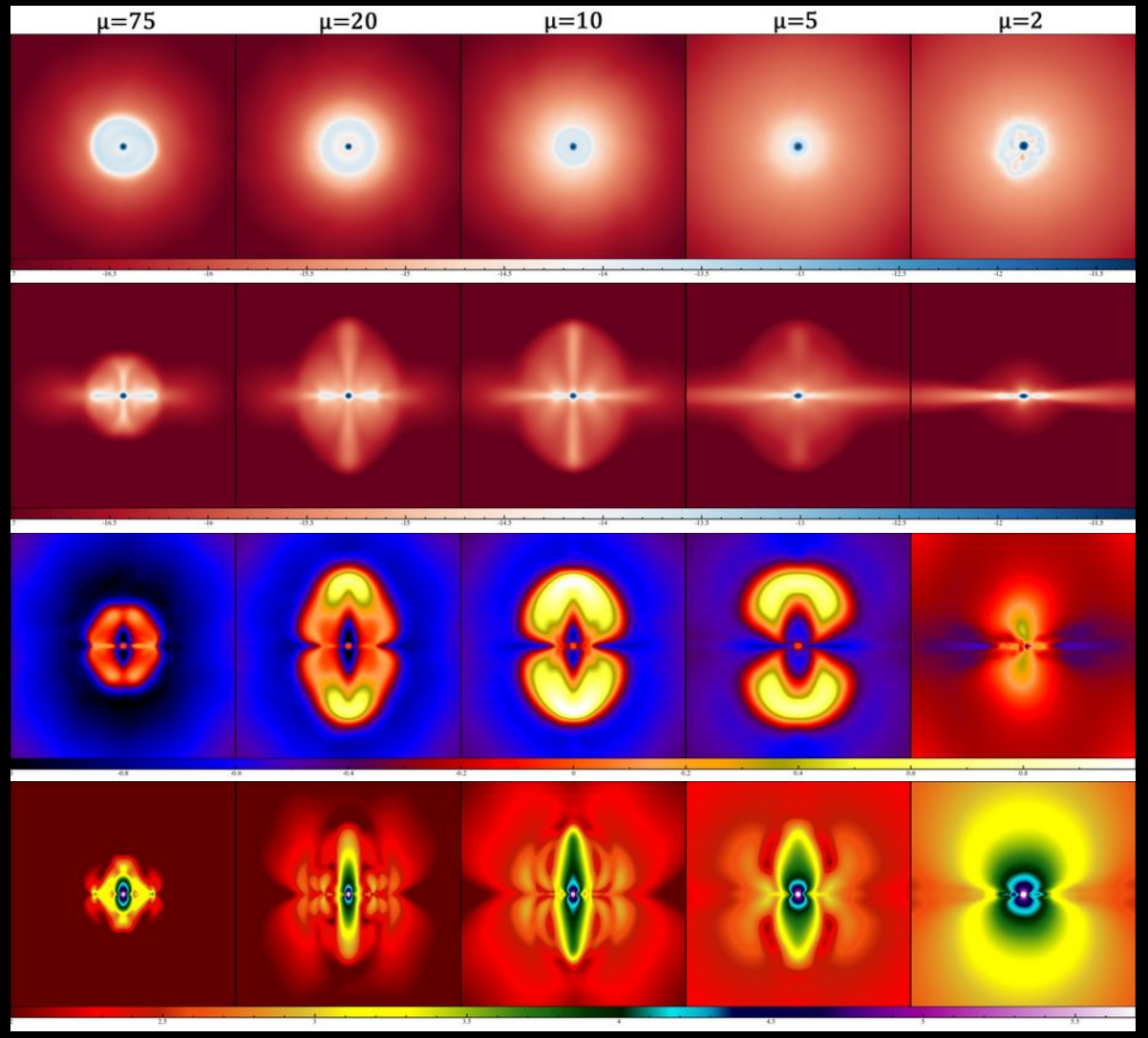


TSPH



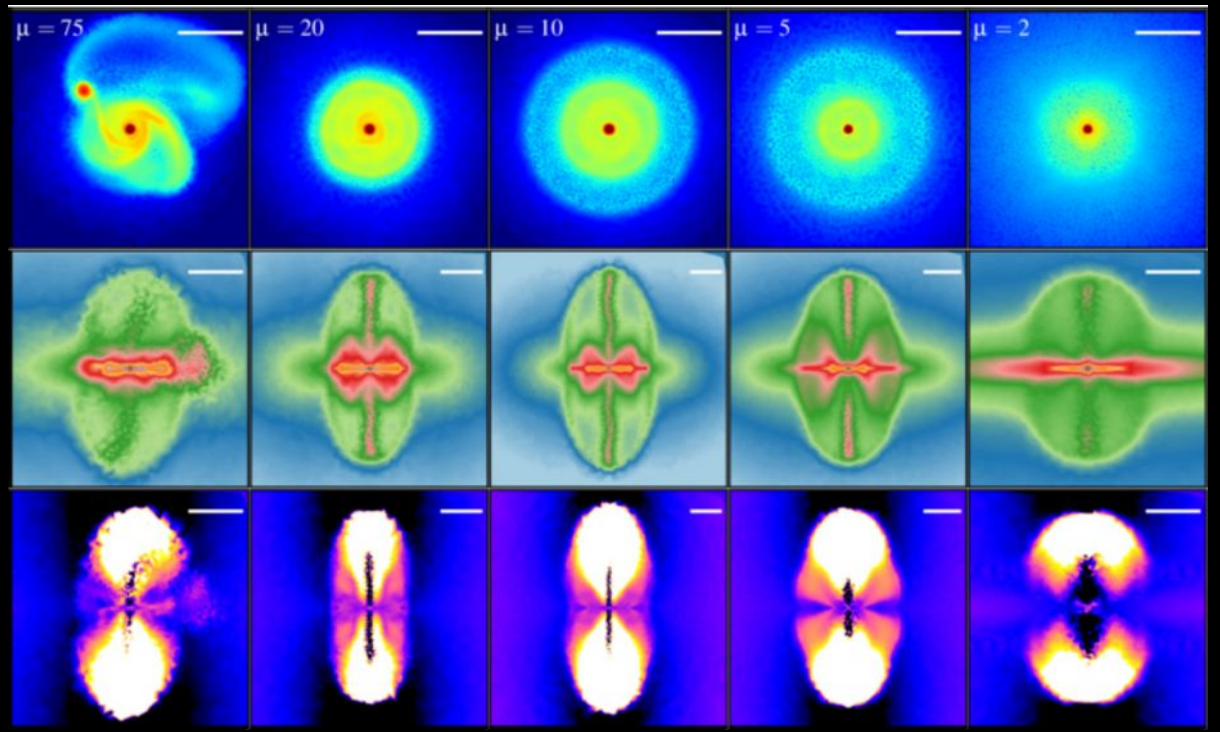
Varying μ – Resolution 50^3

GDSPH

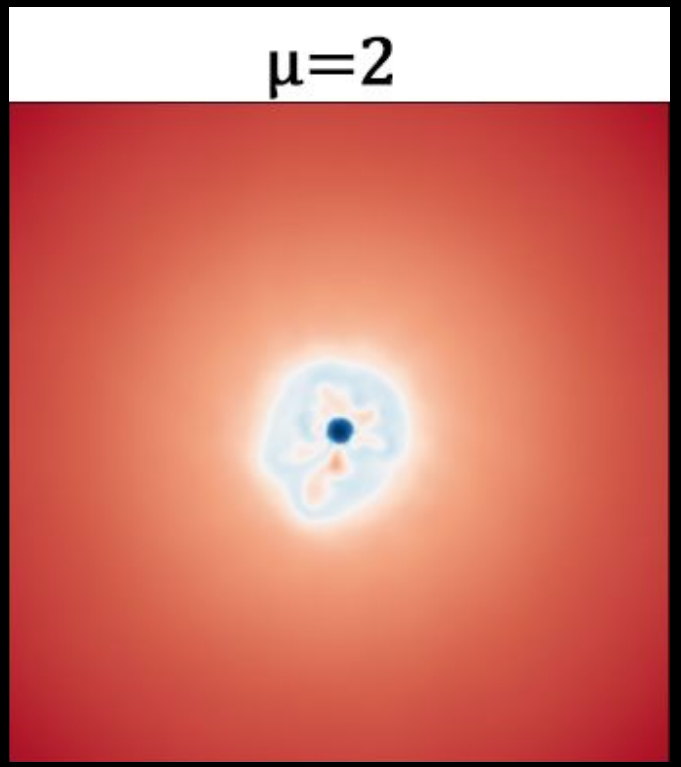


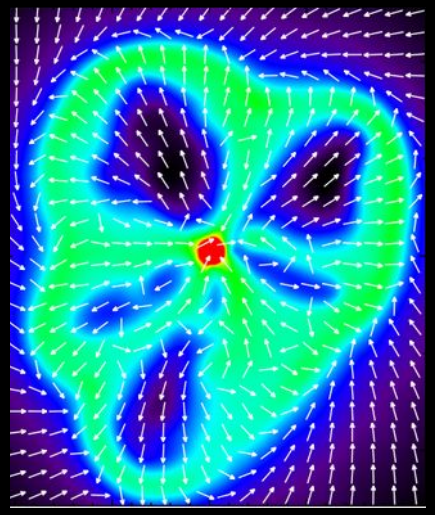
MFM

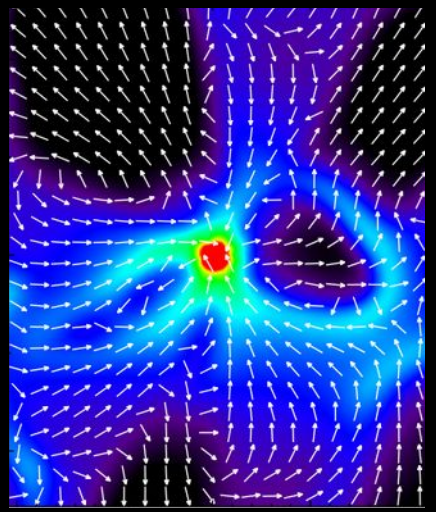
Hopkins & Raives (2016)

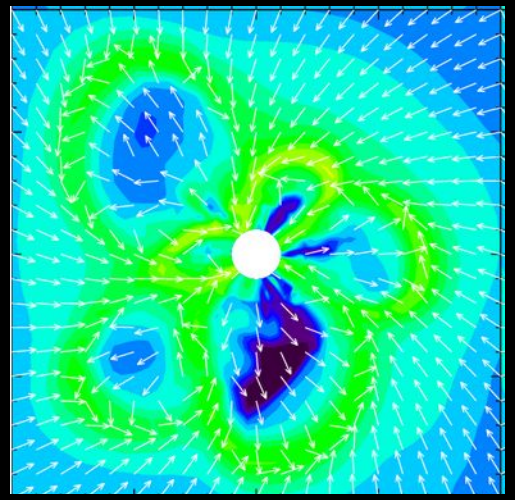


Magnetic Interchange Instability

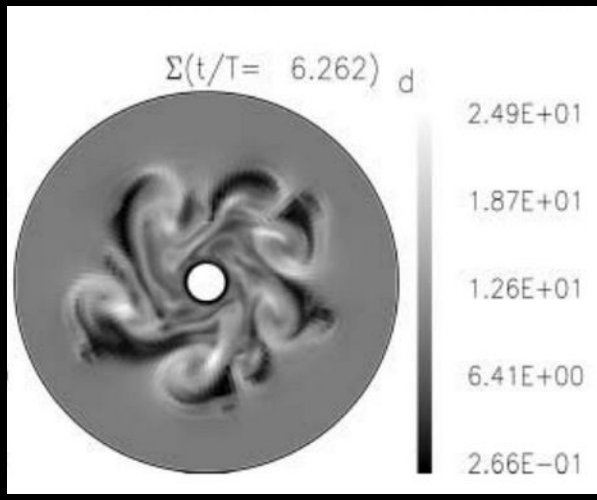








Krasnopolsky (2012)



Kaisig et. al (1992)

Magnetorotational instability (MRI)

- MRI can generate turbulence/stress in accretion disks

Enables efficient transport of angular momentum

- Linearly unstable in ideal MHD (normal mode analysis)
- Secondary instabilities saturate – Non-linear

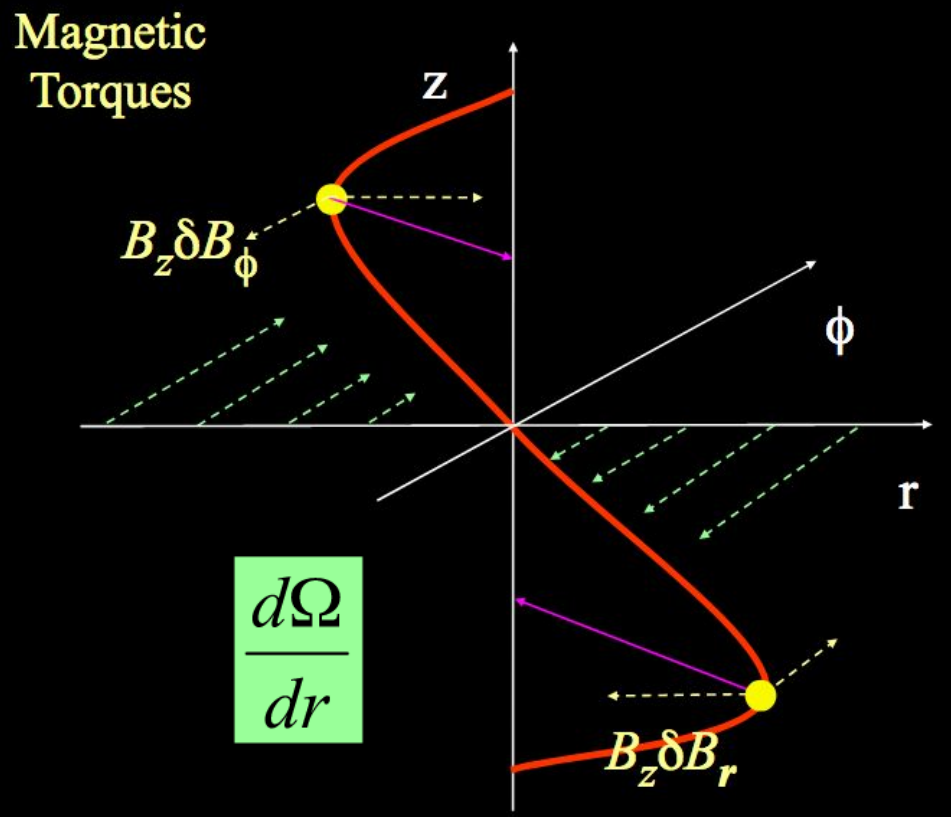
Tearing Instability – driven by ∇B

Kelvin Helmholtz – driven by ∇v

- Diffusion parameters important

$$\mathrm{Re}, Rm, \; Pm = Rm/Re$$

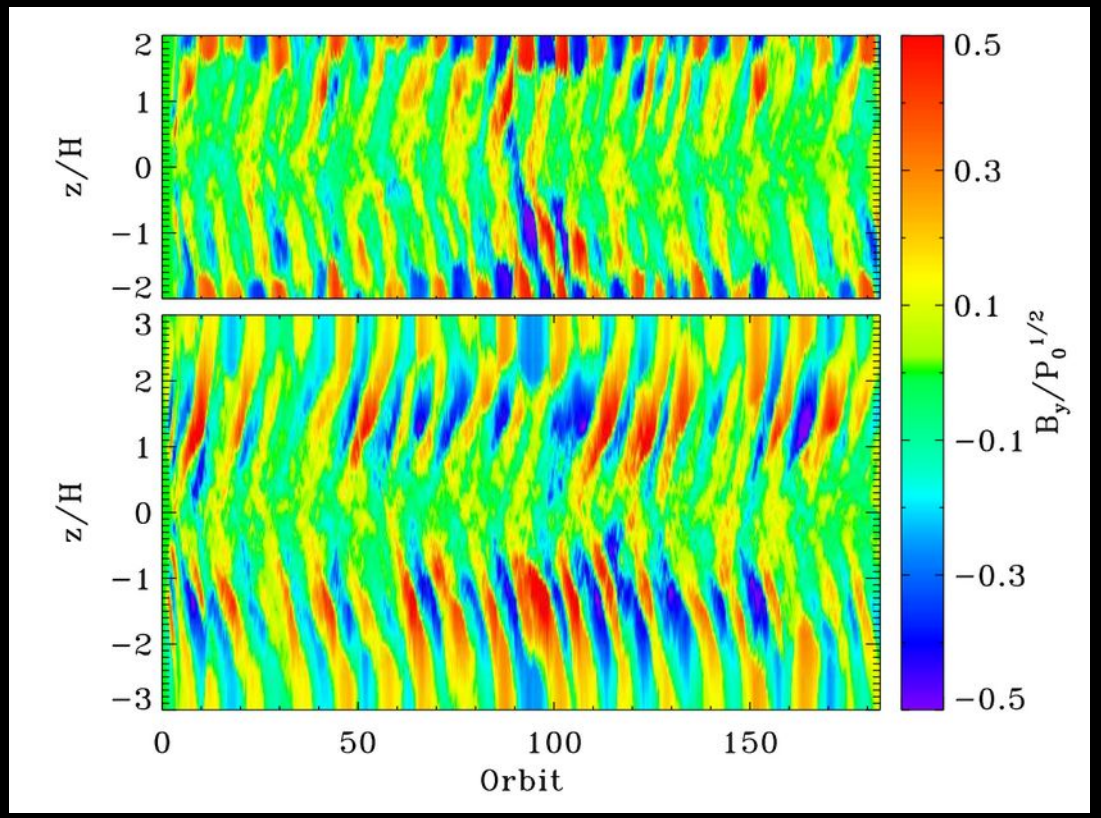
- Non linear evolution studied with local shearing box



Credit: Martin Pessah

MRI in the stratified shearing box

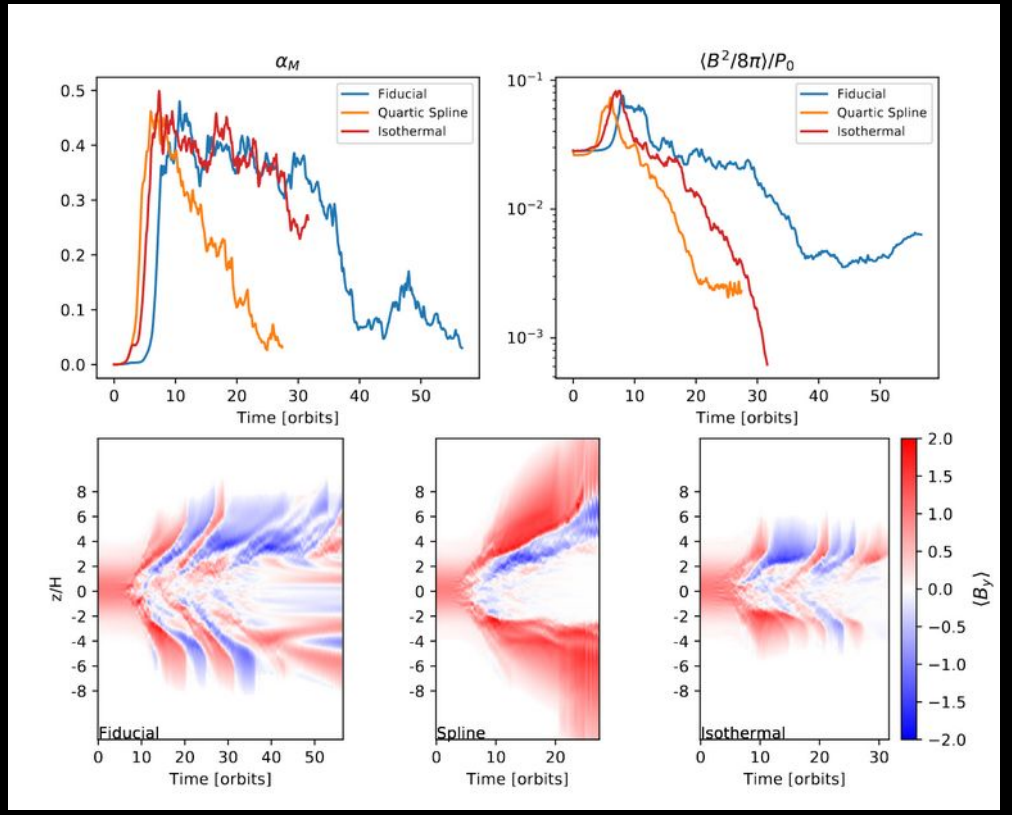
- Case of stratified shearing box
  Zero net magnetic flux
  $\rho = \rho_0 e^{-z^2/2H^2}$ $B = B_0 \sin(2\pi r/H)$
- Cyclic dynamo gives rise to butterfly diagram ---------->



Davis et al. 2010

MRI in the stratified shearing box

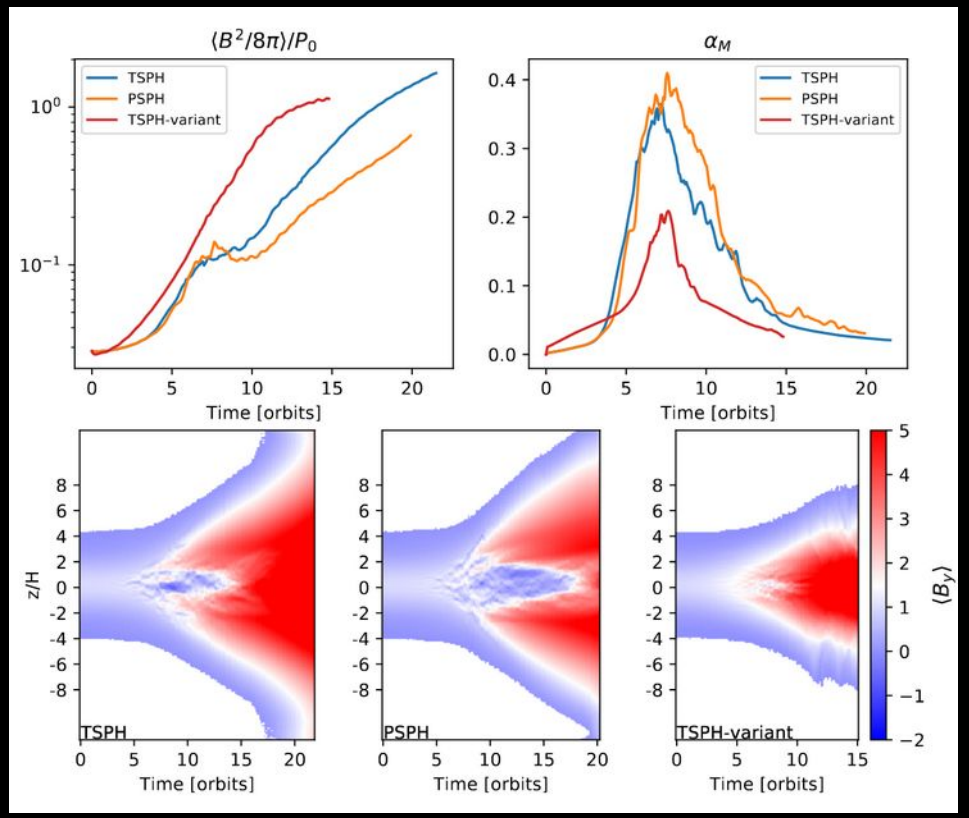
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- MFM method produces butterfly diagram



Deng et al. 2019

MRI in the stratified shearing box

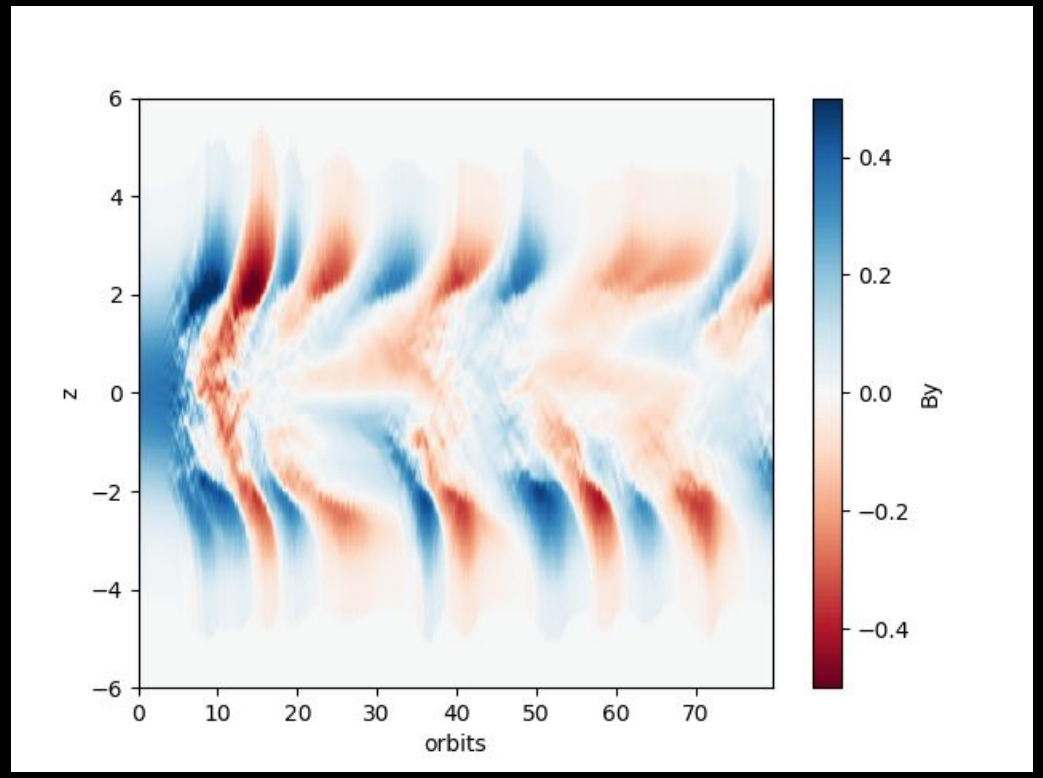
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  Zero net magnetic flux
  $\rho = \rho_0 e^{-z^2/2H^2}$ $B = B_0 \sin(2\pi r/H)$
- Cyclic dynamo gives rise to butterfly diagram ---------->
- MFM method produces butterfly diagram
- SPMHD found to produce unphysical growth in toroidal fields



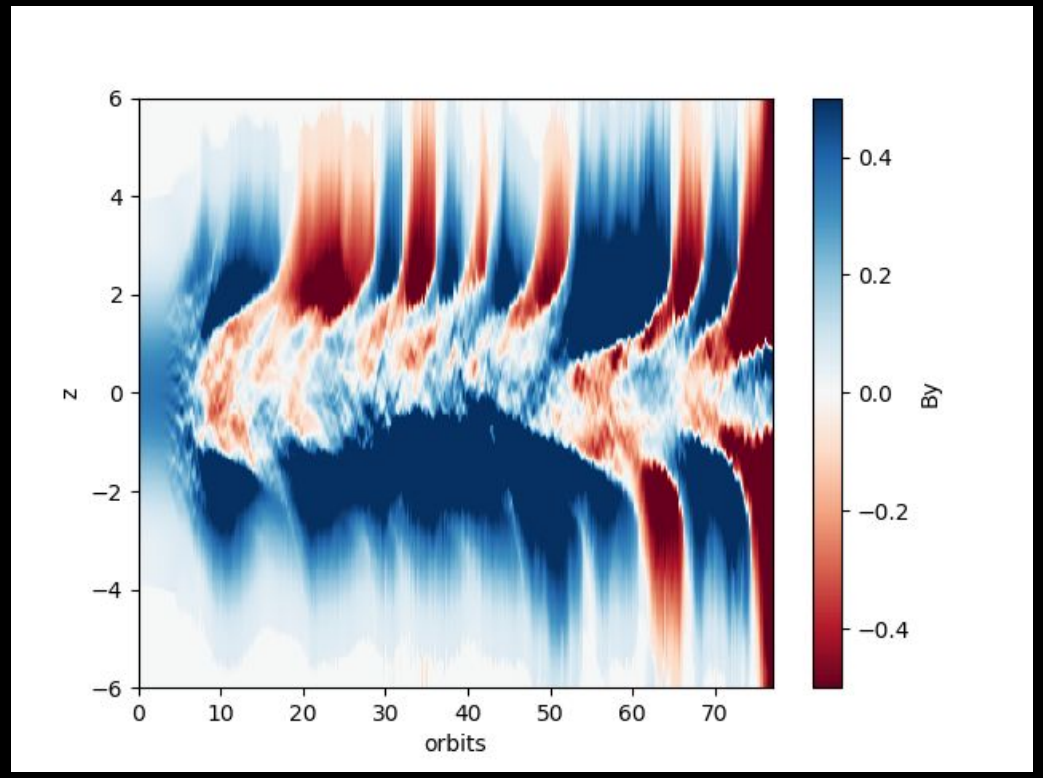
Deng et al. 2019

MRI in the stratified shearing box $\alpha_B = 0.5$

GDSPH

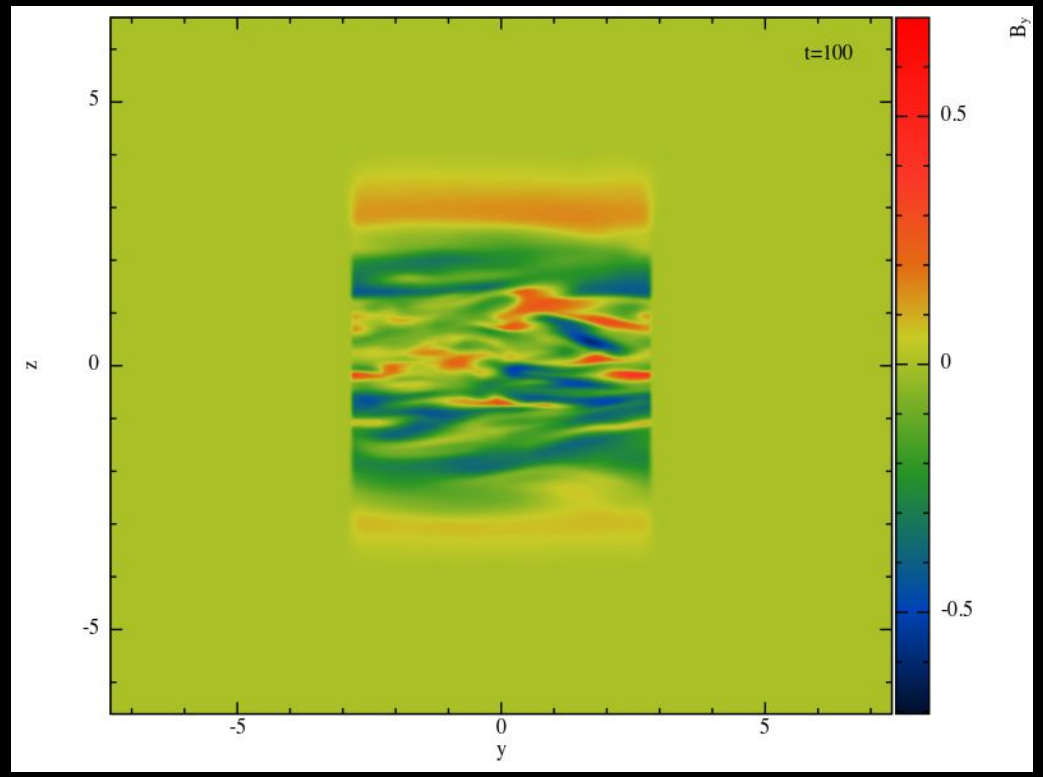


TSPH

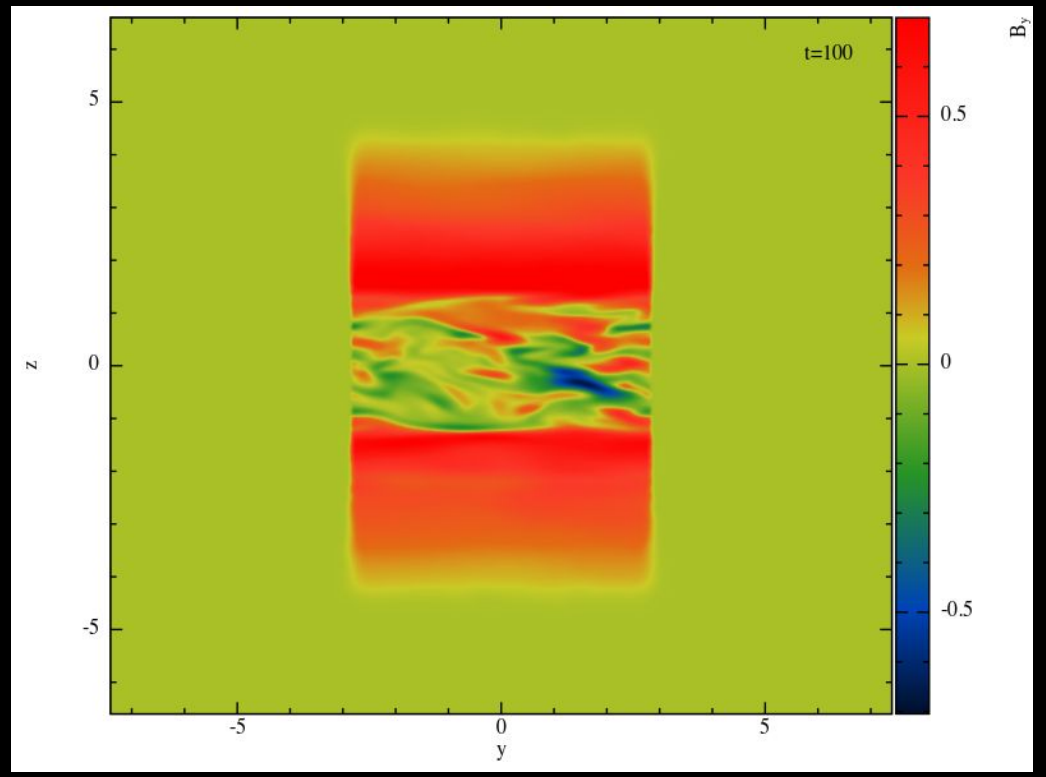


MRI in the stratified shearing box $\alpha_B = 0.5$

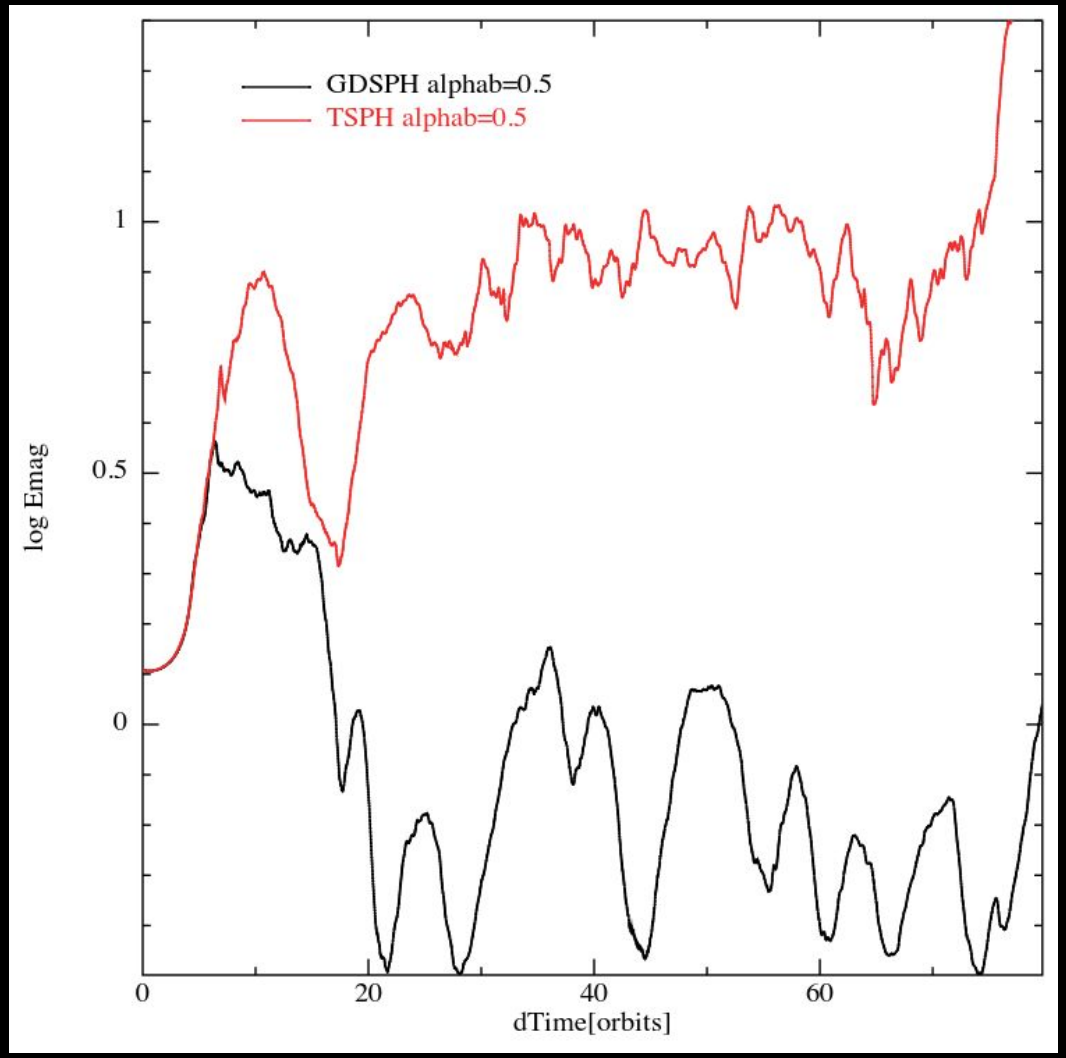
GDSPH



TSPH

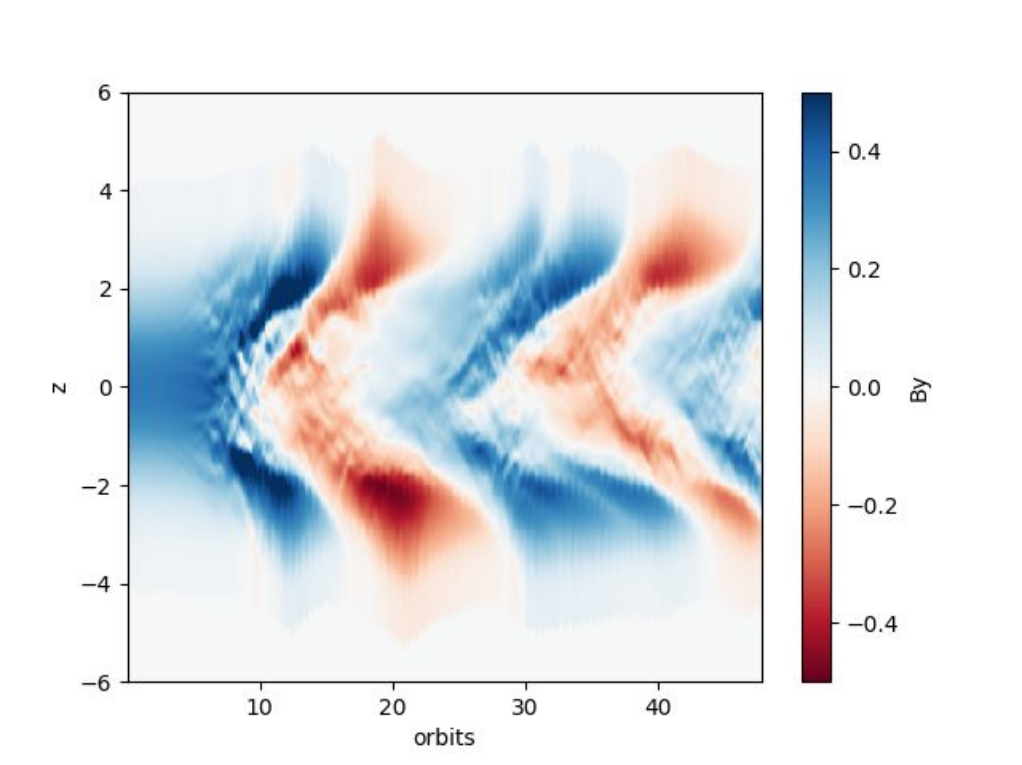


MRI in the stratified shearing box $\alpha_B = 0.5$

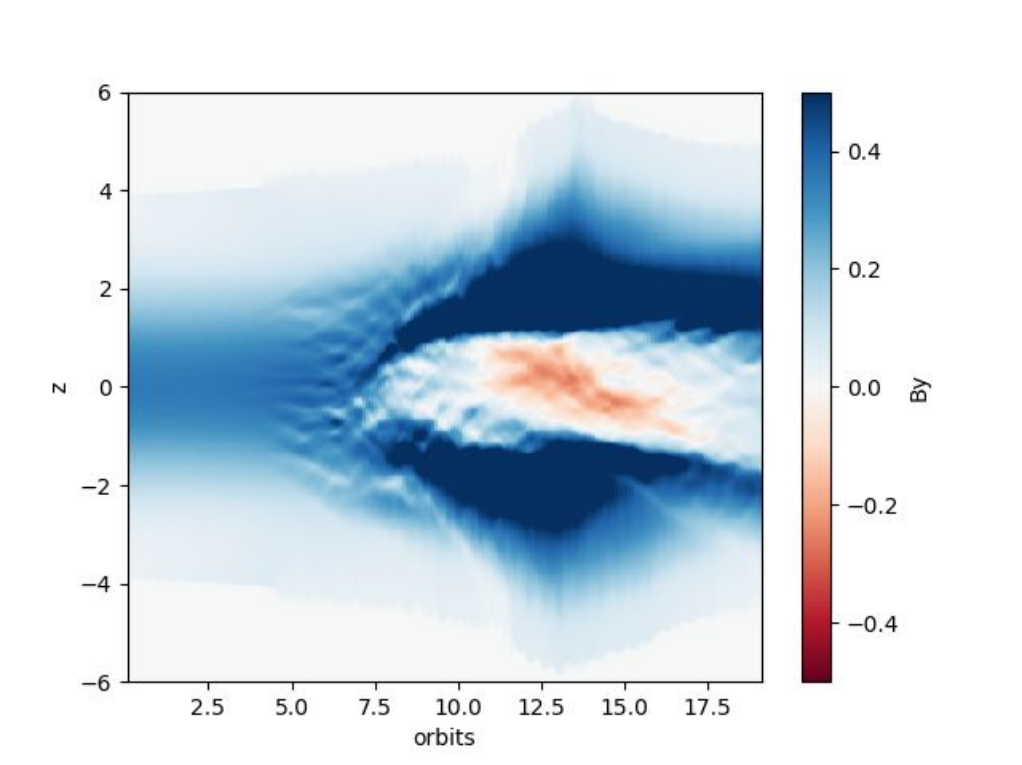


MRI in the stratified shearing box $\alpha_B = 1$

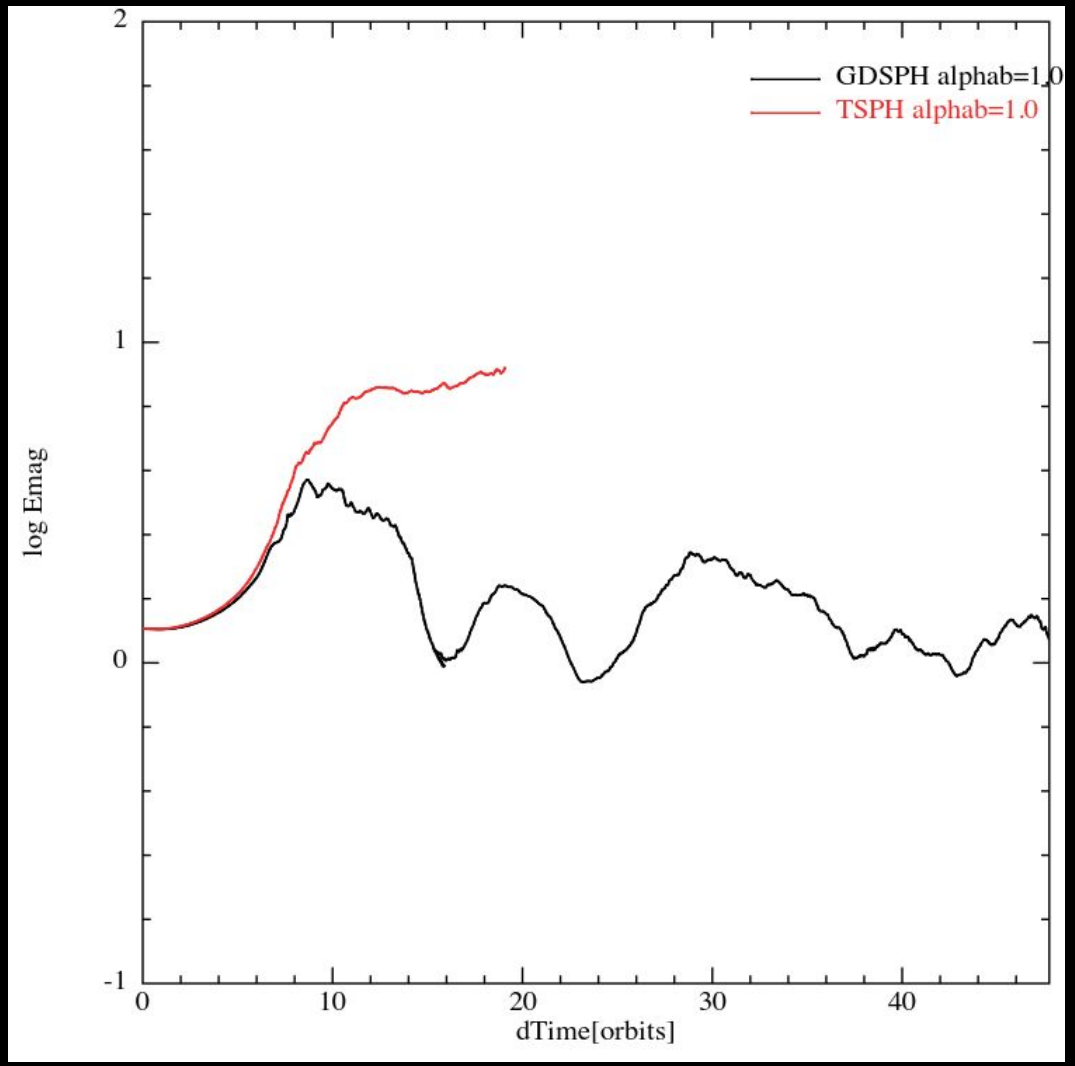
GDSPH



TSPH



MRI in the stratified shearing box $\alpha_B = 1$



Conclusions

- MHD works with GDSPH formalism
- Improves convergence for tests with large density gradients
- No drawbacks compared to TSPH
- Similar to MFM in magnetized core collapse test and the MRI test

Future work

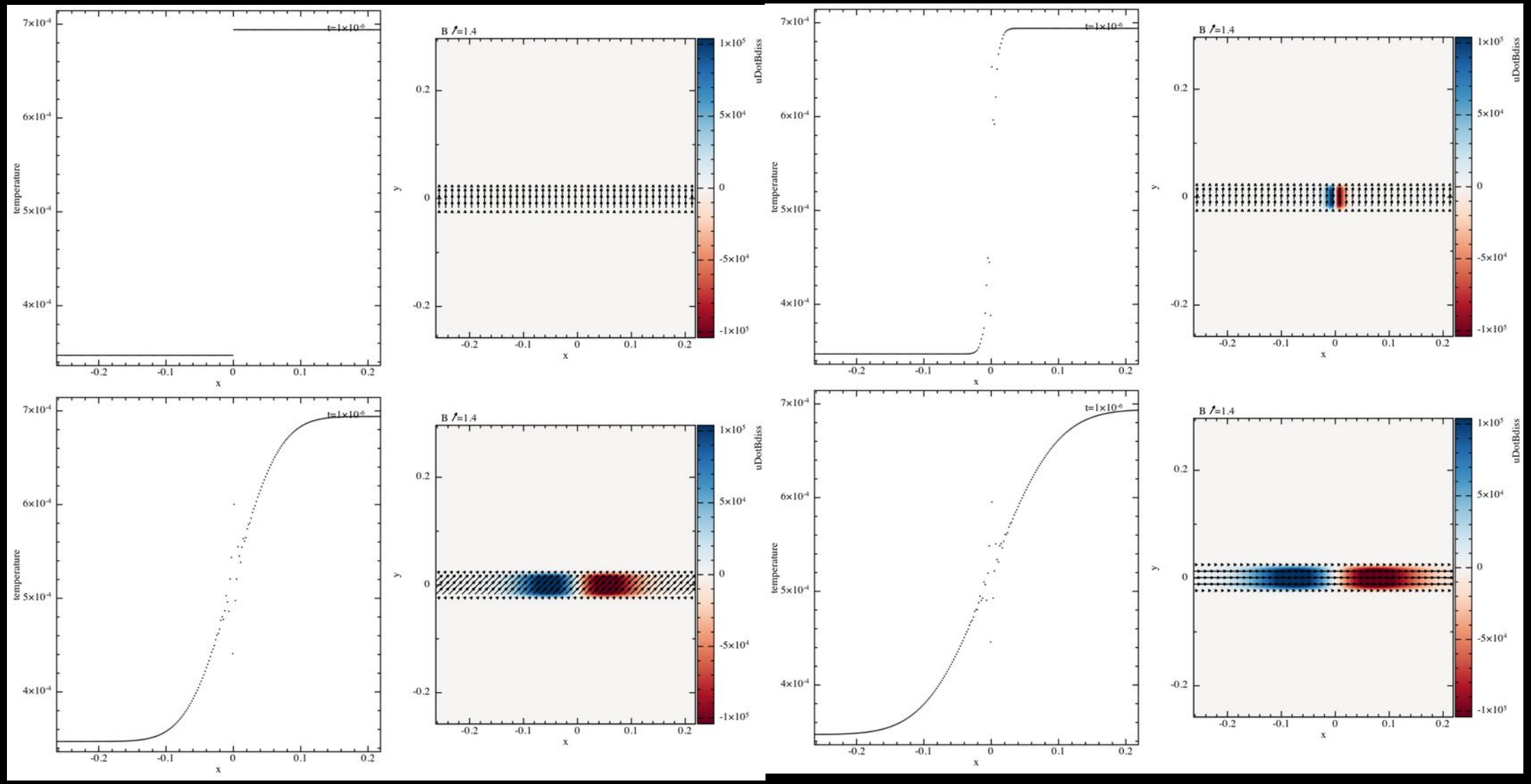
- Galaxy simulation, can we amplify seed fields to equipartition
- Galaxy merger with magnetic fields
- Anisotropic conduction

Can GDSPMHD handle MTI ?

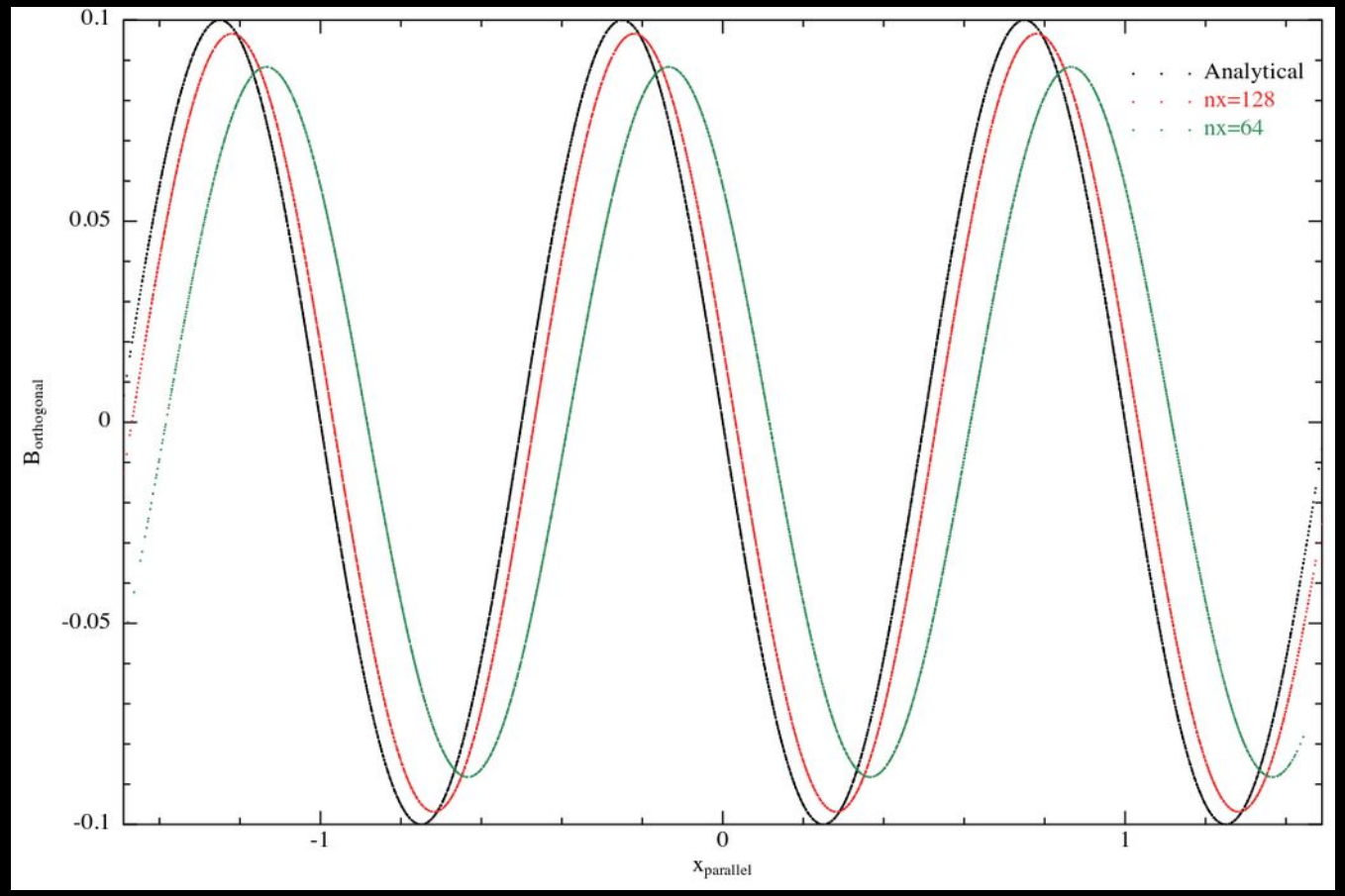
- Non-ideal MHD

Thank you!

Anisotropic conduction

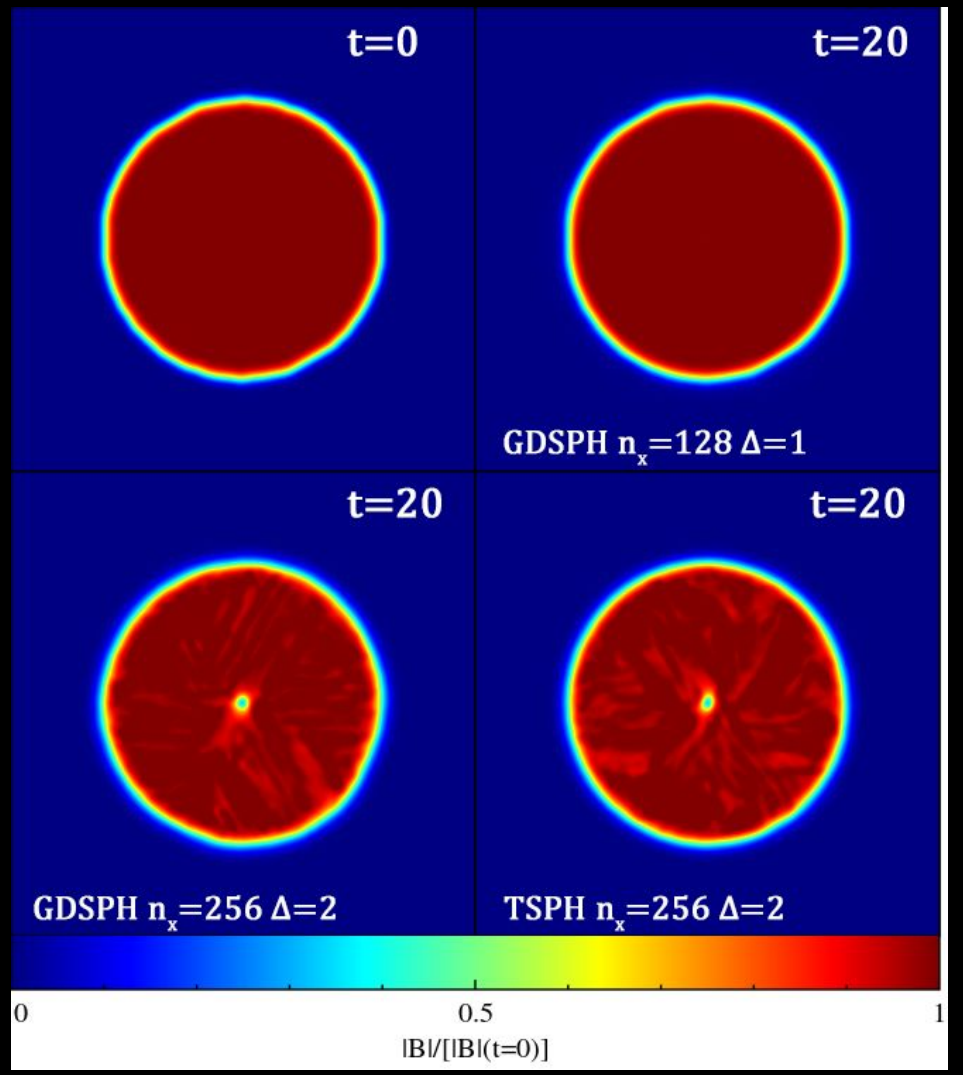


Circularized Alfven Wave

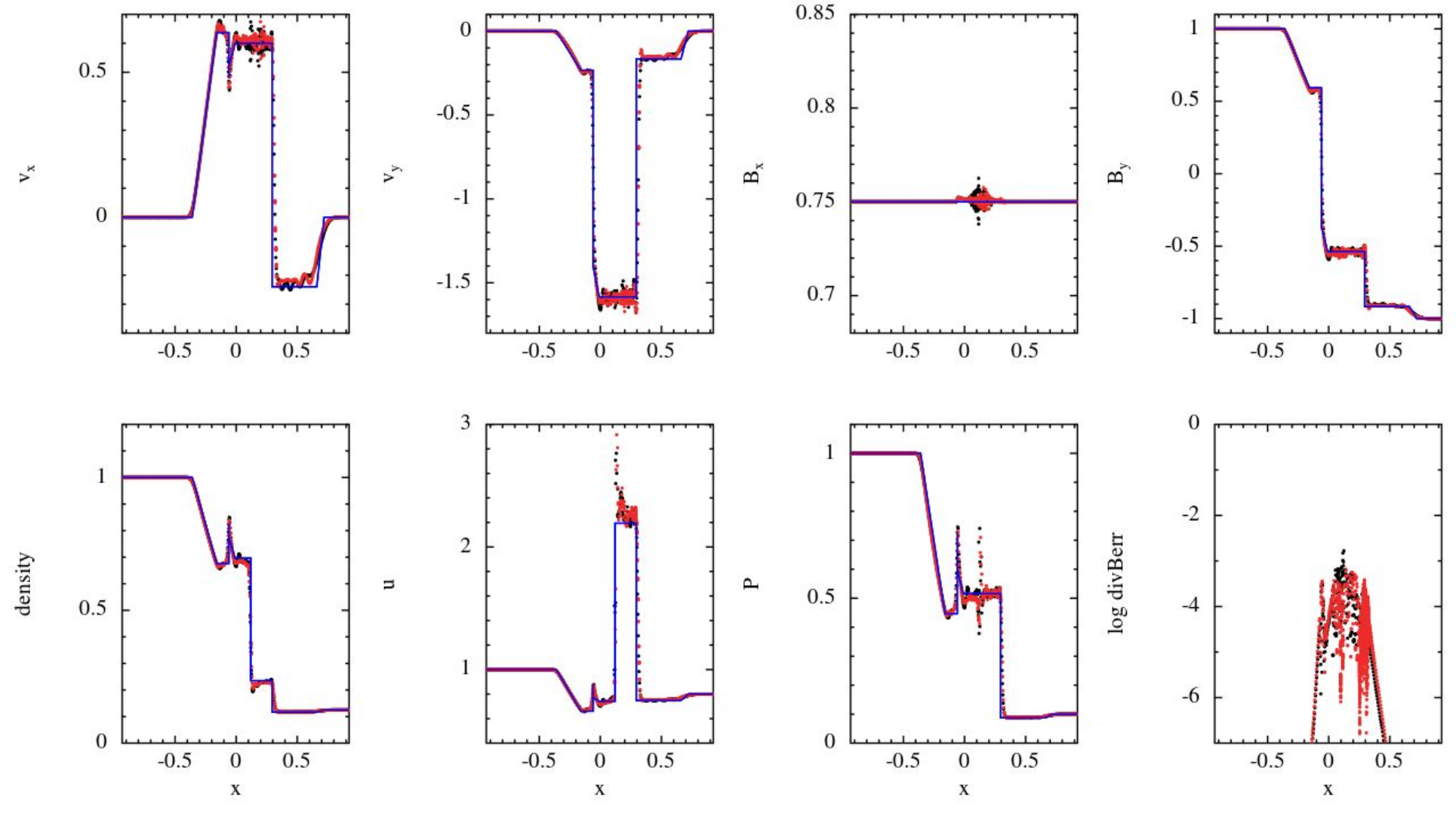


$$L_1 \propto n_x^{-2}$$

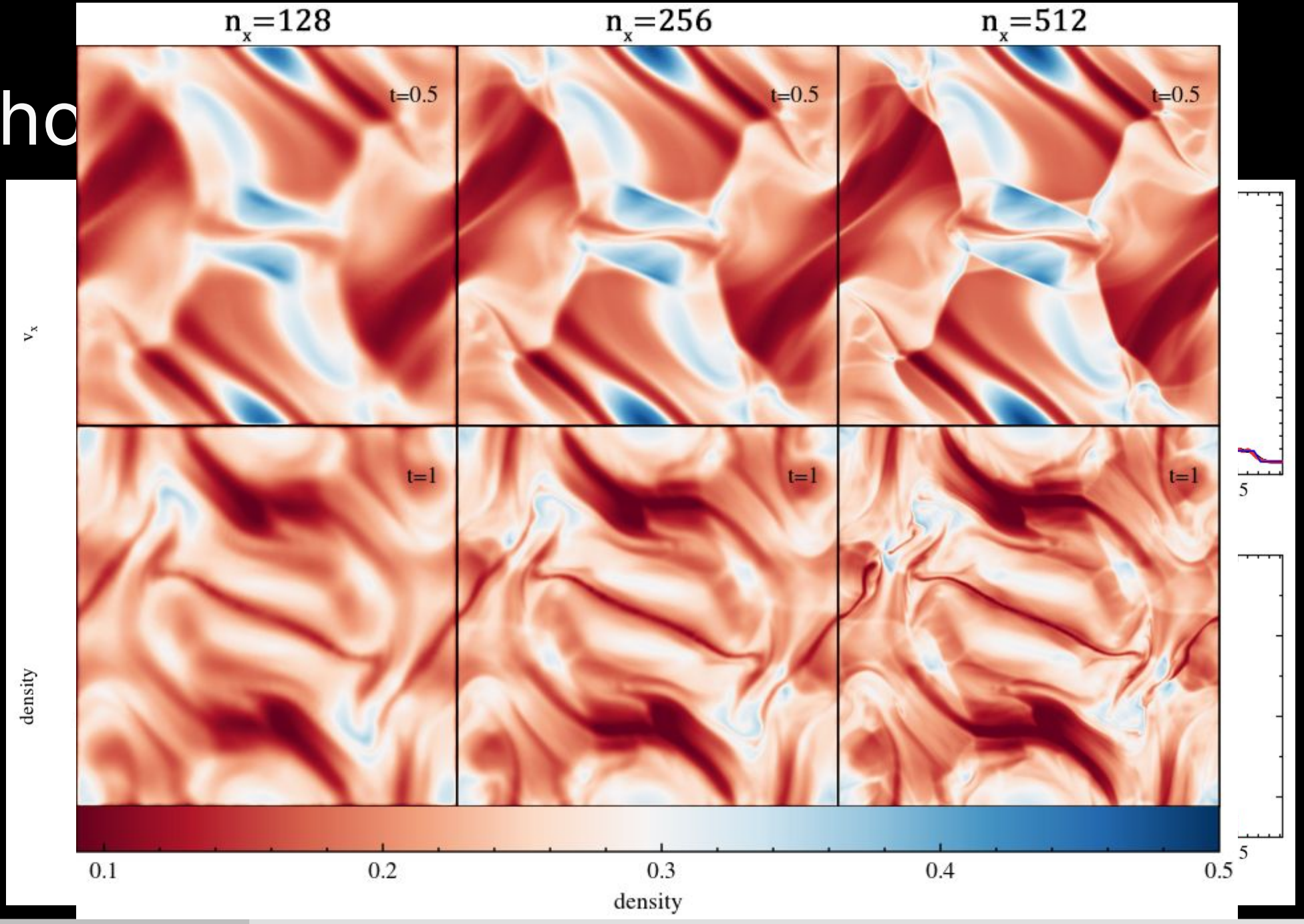
Advection of current loop



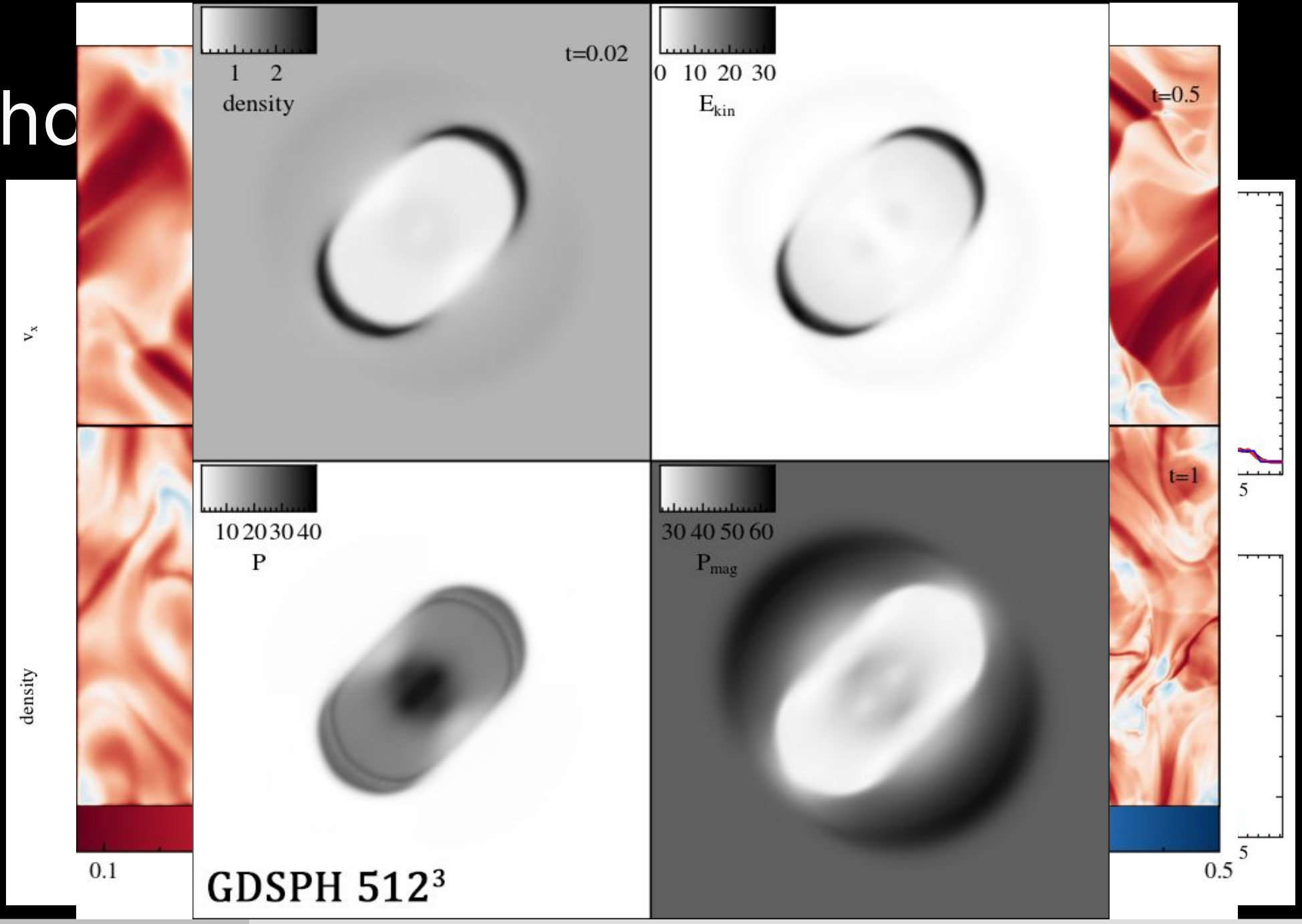
Shock tests



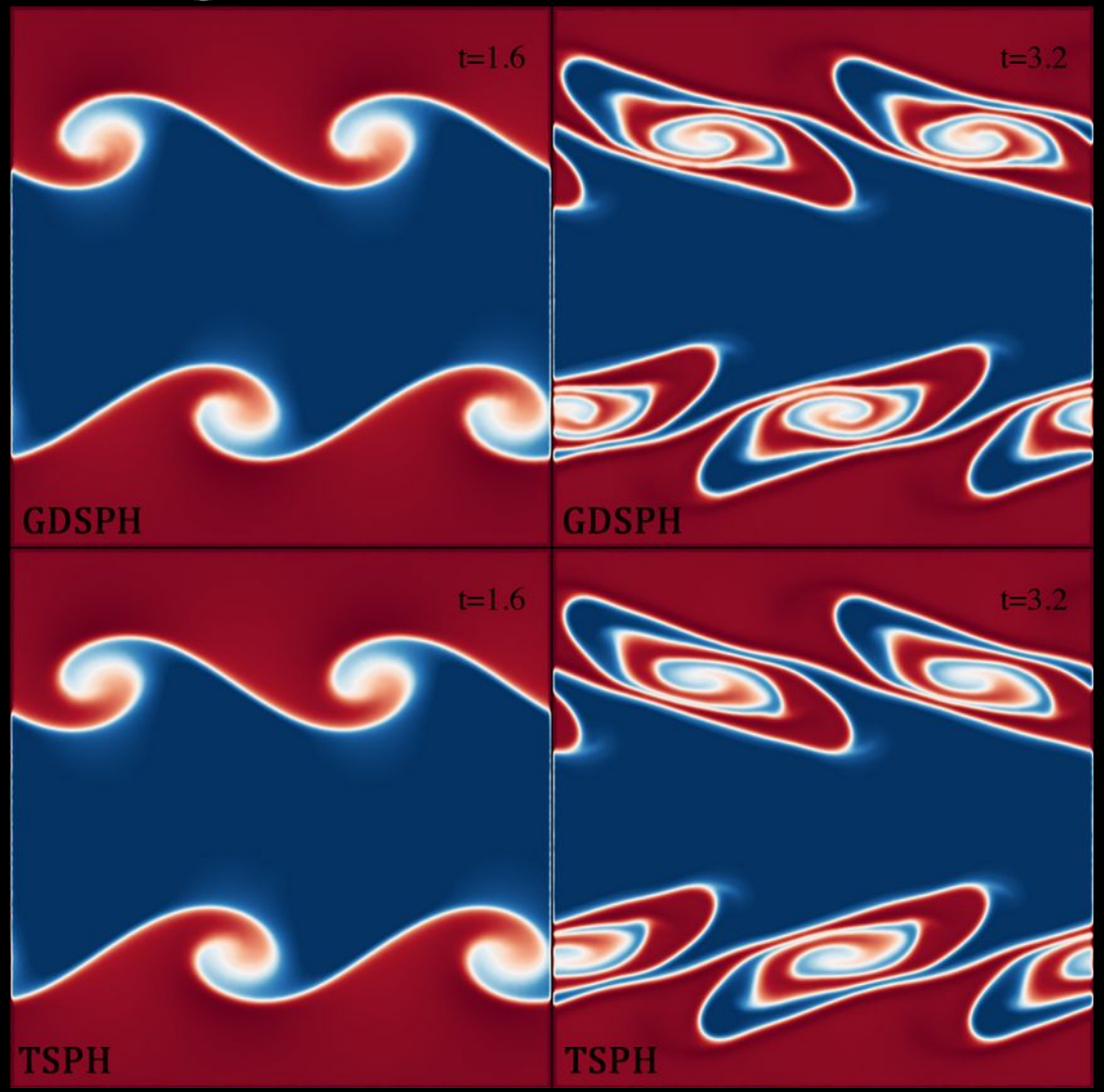
Sho


n_x=128
n_x=256
n_x=512
t=0.5
t=0.5
t=0.5
t=1
t=1
t=1
v_x
density
0.1
0.2
0.3
0.4
0.5
density
5
5

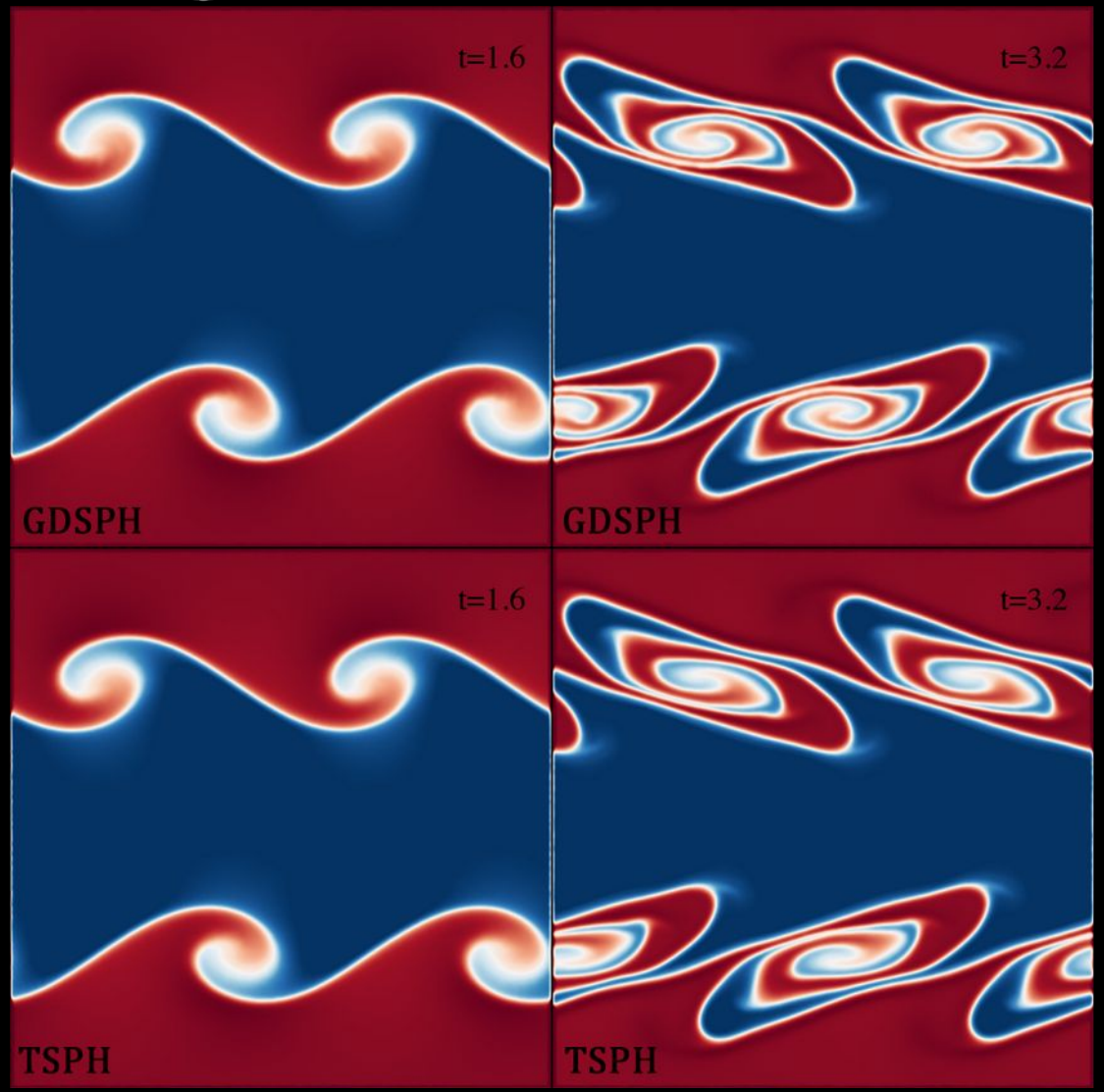
Sho

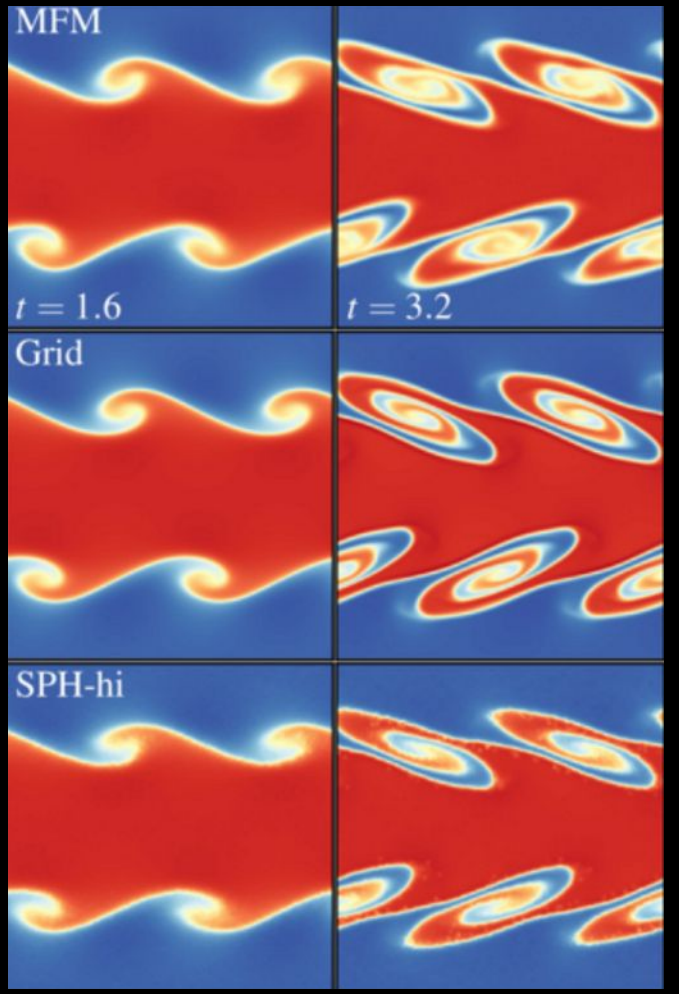

t=0.02
1 2
density
0 10 20 30
E_{kin}
10 20 30 40
P
30 40 50 60
P_{mag}
GDSPH 512³
t=0.5
t=1
v_x
density
0.1
0.5
5
5

Magnetic Kelvin-Helmholtz ($n_x = 256$)

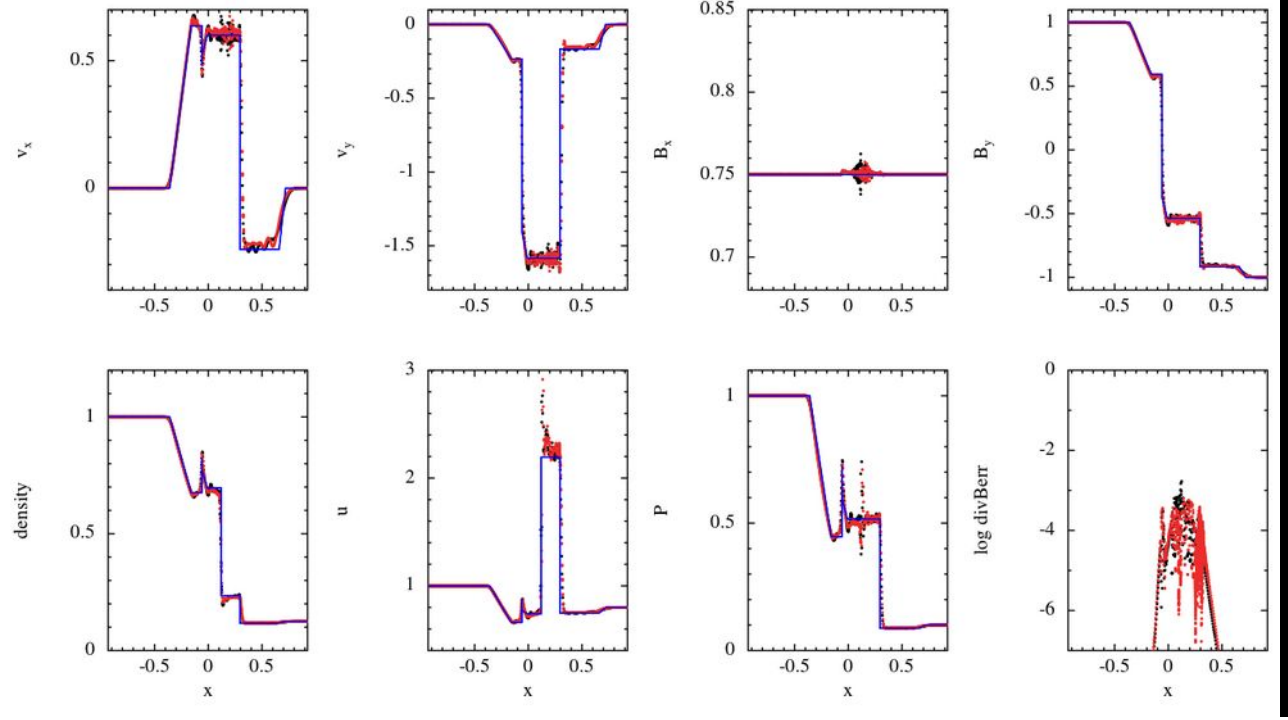


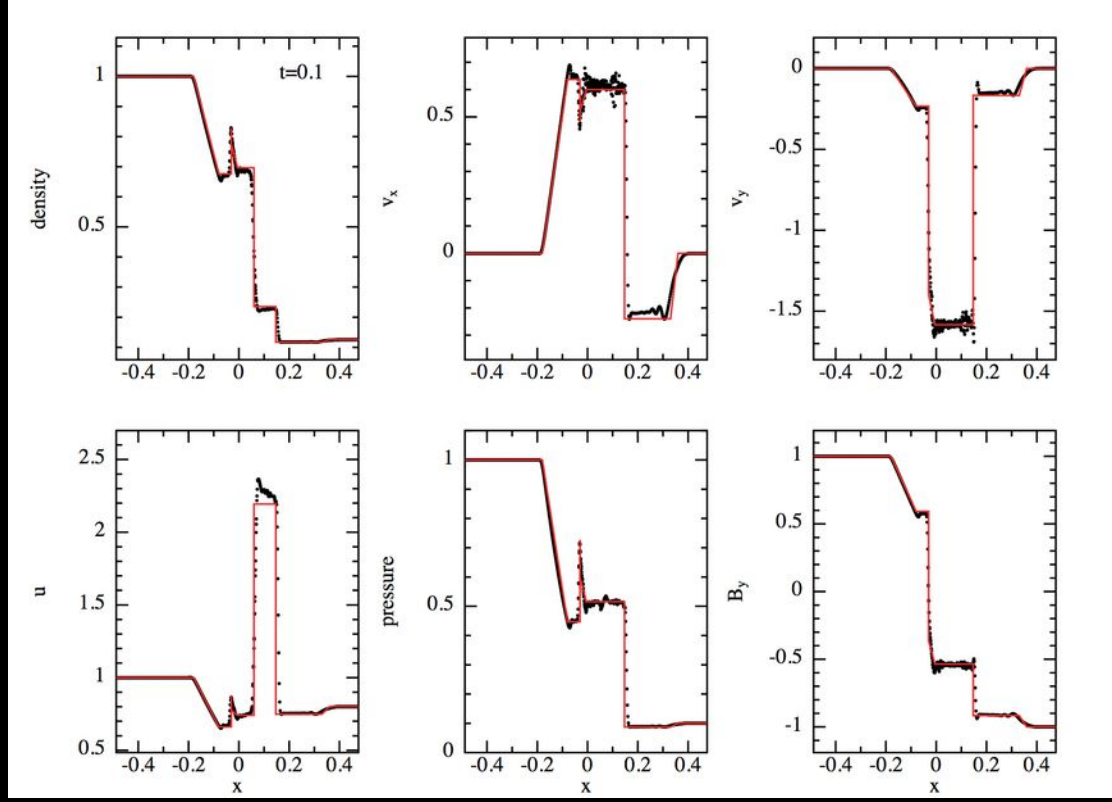
Magnetic Kelvin-Helmholtz ($n_x = 256$)





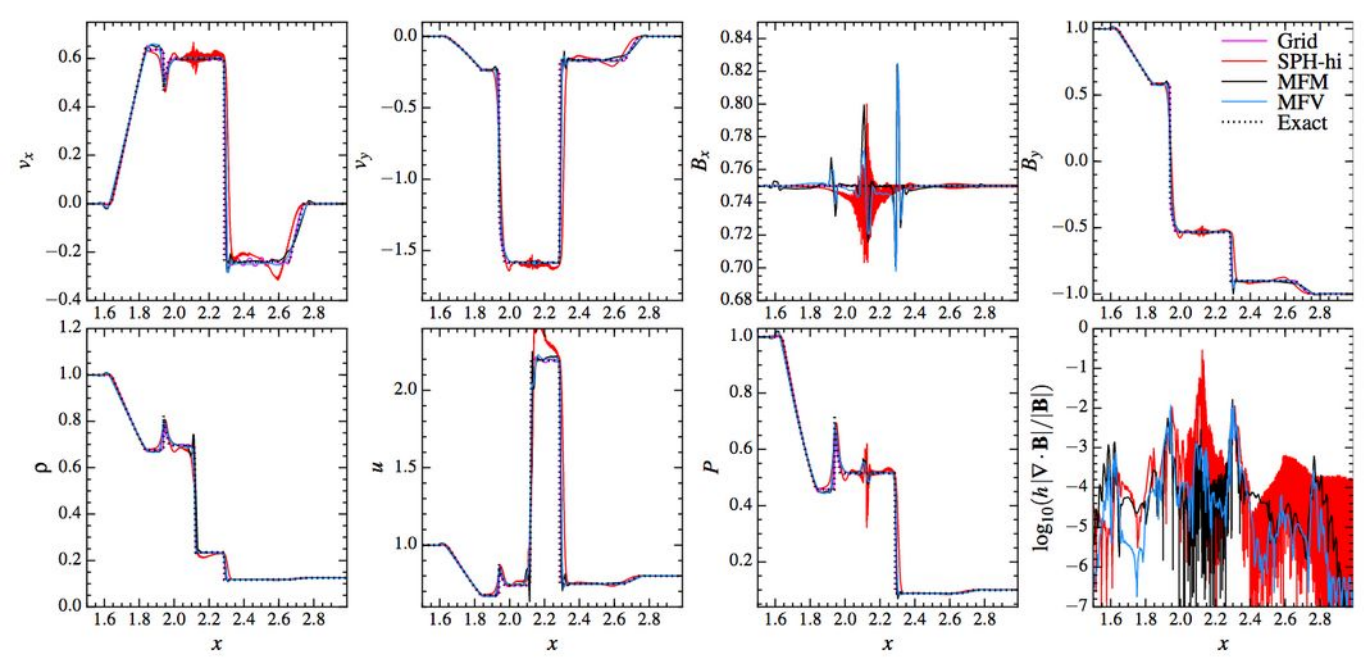
Hopkins & Raives (2016)





PHANTOM

Price et al. (2017)



GIZMO

Hopkins & Raives (2016)